Painter Naming and the Distributional Gap Between Generated Images and Original Paintings

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Abstract

Painter names can improve the proximity of generated images to paintings without recovering their distribution. We study this distinction in 31 interpretable color, spatial and texture coordinates. Exploration covers Monet, Sisley, Pissarro and Cézanne; controlled comparisons use three image services and a separate 70-work Monet/Cézanne panel. Against artist-free controls, naming lowers energy discrepancy in all six original comparisons, with four adjusted rejections; both FLUX effects recur later. A prospective 96-image Cézanne comparison also lowers discrepancy against a generic painting clause on 24 new scenes. A post-result analysis tests global translation and isotropic scaling fitted to generated training outputs. Actual naming is closer than the translation/scale benchmark in all six held-scene averages, but an unchanged translation alone gives lower discrepancy than actual naming for both painters in the later FLUX cohort. Thus even a recurring naming benefit does not uniquely imply recovery of reference variation. Adding scale lowers the repeat-corrected scene-mean prediction residual in all six cell averages, yet worsens reference proximity in five. A further benchmark holds the evaluation mean fixed and finds that the fitted scalar worsens proximity throughout the specified views. Repeat-corrected scene variance contracts, while scene retrieval changes in both directions. Both palette collections leave named-minus-generic differences in chroma response unresolved. Processing challenges and common-square sensitivity delimit the measurement interpretation. The findings distinguish reference proximity from conditional organization in digital feature space; they do not identify a perceptual or internal generative mechanism.

1 Introduction

A generated painting may evoke a painter while the generated collection differs from that painter's works. Familiar motifs and palettes can accompany a narrow range of compositions or repeated surface patterns. Conversely, broad output variation can lie far from the intended paintings. The research question is therefore how a prompting intervention changes a distribution, beyond distance to an average feature vector.

We examine adding a painter's name to otherwise unchanged scene prompts. The reference target is an explicit panel of digital painting reproductions, measured in fixed color, spatial and texture coordinates. Energy discrepancy incorporates both cross-collection distances and within-collection variation. A lower value indicates improved proximity in that geometry, but leaves an explanatory question: which aspects of the generated collection have moved toward the references?

Our diagnostic compares actual named outputs with generated-only translation and translation/scaling benchmarks. Fitting excludes reference objectives; evaluation holds out whole scenes or applies original maps unchanged to a later collection. The transformed coordinates are diagnostic objects, not proposed image edits.

Four-painter exploration provides context for the gap. Controlled collections then compare Monet and Cézanne on three services using 24 repeated scenes. Actual naming beats translation/scaling in all six primary held-scene averages, but translation alone is closer to the references in the later FLUX cohort. Corrected variance, held-repetition retrieval, palette interventions and processing challenges test distinct interpretations of these proximity gains. A prospective Cézanne comparison extends the proximity result to a generic painting clause on 24 newly specified scene briefs.

The contribution is an empirical test of what a painter-name proximity gain can and cannot establish in a repeated-prompt design. Conditional evaluation and location/scale transformations are established tools; the evidence lies in their observed successes and failures across the painting comparisons. We retain the four-painter descriptive findings, distinguish prespecified tests from subsequent diagnostics, and provide computations from the recorded numerical observations.

2 Related work

Color, image structure and information-theoretic measures have been used to study variation across painting collections (Kim et al., 2014; Sigaki et al., 2018; Lee et al., 2020). Kim et al. (2026) contrasted autoencoder and CLIP representations and investigated context-keyword image-to-image transformations in art-historical development. Their contextual prompts omit artist names and style-period labels. Our text-only intervention explicitly adds a painter's name and compares generated distributions with fixed reference panels, without supplying an original composition as input.

Deliège et al. (2025) combine three experts' assessments with quantitative measures of relative shift, dispersion and overlap for five movements and ten painters. Their generated sample contains 300 Midjourney v6 images, whereas historical reference ranges come from the experts' knowledge of artistic production, without presenting a fixed historical-image corpus. Our references are explicit panels of digital reproductions, and our repeated scene prompts support a within-service naming contrast. The studies therefore use different reference targets and measurements: expert characterizations of style versus observed image features.

AI-Pastiche evaluates artistic imitation through a public authenticity survey and prompt-adherence assessments by the research team and additional volunteers (Asperti et al., 2025). AI-WikiArt conditions generation on captions of original paintings and evaluates attribution with vision-language models (Fu et al., 2025). These studies address imitation and attribution under different conditioning and judgment procedures. Our comparison holds scene text fixed across naming conditions, linking distributional changes to a specified prompt intervention.

Su et al. (2025) benchmark prompted-artist recognition using fixed-content artist substitutions and held-out prompts. They also compare named/free images and real-artist prototypes in CLIP space across prompt complexities. Our contribution is therefore not the name-substitution control or unseen-prompt evaluation itself. We study full-distribution proximity and repeated-scene organization, testing the behavior of generated-only moment maps against both targets.

Asperti (2026) reports human/generated separation in CLIP space and probes it with interpretable descriptors, multiscale representations and image inversion. Their crop, resolution and encoding probes motivate our processing sensitivities; visually small inversion changes despite embedding displacement also caution against interpreting computational separation as perceptual difference. We examine a prompting intervention.

Conditional evaluation already distinguishes variation within conditions from the organization of condition means (Benny et al., 2021). In text-to-image models, increased prompt complexity can reduce conditional diversity while improving reference-based distribution measures (Zhang et al., 2025). Our intervention holds scene wording fixed and adds a painter clause; the held-scene transformation benchmark tests what predicts its measured effect. Neighborhood coverage can also respond asymmetrically to inward and outward changes in high-dimensional distributions (Khayatkhoei and AbdAlmageed, 2023). That result does not establish a distortion in these 31 coordinates, but motivates comparing coverage gains with those produced by the same fitted transformations.

Generative-model evaluation distinguishes fidelity, diversity and coverage (Naeem et al., 2020), and preprocessing can materially alter feature comparisons (Parmar et al., 2022). We apply energy discrepancy (Székely and Rizzo, 2013) alongside spread and neighborhood measures. Compositional benchmarks evaluate attribute binding and object relations, including color binding (Huang et al., 2023). Our palette experiment measures global median chroma, a different aspect of instruction response from object-specific color assignment.

3 Study overview and feature representation

The analyses share a fixed image representation but use separate cohorts and comparisons (Table 1). Throughout, *reference proximity* means lower energy discrepancy, *spread* means total feature variance, and *contraction* means lower spread under named than artist-free prompting. These quantities describe digital image distributions.

Each image is measured in 31 interpretable coordinates: 11 color, eight spatial and 12 digital-texture features (Appendix A.1). The primary pipeline applies orientation metadata, converts valid embedded color profiles to sRGB, assumes sRGB for compatible unprofiled files and preserves aspect ratio at a 512-pixel short side. Images are opaque and resized with Lanczos filtering, without upsampling or discretionary cropping. Each coordinate is standardized by its median and interquartile range (IQR) in a separate development collection of 221 works: 101 Monet, 36 Sisley, 48 Pissarro and 36 Cézanne, with equal total weight per painter. Reference and generated images are excluded from fitting this transformation. The scaler therefore uses separate works by the same painters, rather than unseen painter identities.

For reference vectors $X = \{x_i\}$ and generated vectors $Y = \{y_j\}$, let a_i, b_j be nonnegative weights summing to one within each set. We use the empirical V-statistic energy discrepancy

$$\widehat{\mathcal{E}}(X,Y) = 2\sum_{i,j} a_i b_j \|x_i - y_j\|_2 - \sum_{i,k} a_i a_k \|x_i - x_k\|_2 - \sum_{j,l} b_j b_l \|y_j - y_l\|_2. \quad (1)$$

This statistic incorporates distances within and between the two collections. Its finite-sample baseline need not be zero for independent samples from the same population. We interpret lower values as closer empirical distributions in the specified feature geometry, not as a calibrated degree of artistic similarity.

4 Exploratory comparison across four painters

4.1 Corpus and descriptive methods

The exploratory corpus comprises 297 Monet, 106 Sisley, 141 Pissarro and 105 Cézanne digital reproductions, selected through recorded outdoor-place metadata. Two requested OAuth aliases, `gpt-image-1` and `gpt-image-2`, each supply 64 images per painter and prompt method: 16 scene templates repeated four times. The aliases are request labels, not independently attested model snapshots. Three formulations name the painter: a by-name prompt; an explicit style instruction appended to the artist-free scene; and that instruction plus attention to color relationships, organization of forms and painted marks. This gives 1,536 painter-conditioned outputs. A further 384 artist-free outputs provide method-specific controls shared across painter comparisons. Appendix A.4 gives the exact construction of the three named/control contrasts. No original painting is supplied to generation.

Analysis	Data and comparison	Evidential role
Four-painter exploration	649 references; 1,536 painter-conditioned and 384 artist-free outputs from 2 requested aliases, 3 prompt methods and 16 repeated scenes	Post-hoc distribution, spread, grouped discrimination and matched-size reference comparisons; separate from the two controlled cohorts.
Study 1 primary	864 detailed outputs: 3 services $\times$ 2 painters $\times$ 24 scenes $\times$ 3 repeats $\times$ 2 conditions; 142 additional short-scene outputs	Eight prespecified paired energy tests: six named/free and two detailed/short-scene named contrasts.
Study 1 descriptive	Same images; 38 Monet and 32 Cézanne references	Prospective spread, PCA, detection, coverage and painter-distance summaries; subsequent alignment double contrasts, matched-real coverage, decomposition, retrieval and extended sensitivities are post-result.
Study 2 primary	192 new outputs: 1 service $\times$ 6 fixed scenes $\times$ 4 repeats $\times$ 4 arms $\times$ 2 palettes	Two prespecified named-minus-generic chroma-response interactions; shared free/generic controls and secondary nominal contrasts.
Measurement follow-up	All 70 Study 1 references in 10 fixed conditions; central-square views of all 1,006 Study 1 outputs	Computational response checks and post-result window sensitivity; no new reference or generated sample.
Temporal follow-up	72 new FLUX naming outputs and 192 new OAuth palette outputs	Separate prospective collection; four primary endpoints with a new Holm family; fixed templates and previously exposed references.
Moment-map diagnostic	Retained 864 detailed Study 1 and 72 later FLUX outputs	Post-result whole-scene prediction, fixed-map temporal transfer, subsequent evaluation centering and finite-repeat correction; no additional hypothesis-test family.
Generic-clause follow-up	211 of 288 original outputs retained; separate 96-output Cézanne/generic cohort on 24 new scenes	Original comparisons unavailable; one prospectively fixed 48-pair successor test at $\alpha = .025$, without pooling or map evaluation.

Table 1: Design and evidence hierarchy. Exploratory collection: 5–6 September 2026 UTC, including two later retries. Study 1 collection: 6–7 September 2026 UTC; Study 2: 8 September 2026 UTC; temporal follow-up: 9–10 September 2026 UTC; generic-clause follow-up: 10 September 2026 UTC. Painter selection followed earlier exploration, while both studies' primary contrasts were fixed before their generation. Sensitivity views reuse images; they are not independent replications.

Two initially refused requests, one for Pissarro and one for Cézanne, succeeded in later trials. These outputs complete a 1,920-image descriptive grid, but their later timing prevents the planned randomization inference for the original full grid. All analyses in this section were developed after the data were observed.

For visualization, each painter has one weighted principal-component analysis (PCA) basis fitted to its originals and all six alias/method groups. Originals carry half the total mass, and each generated group carries one twelfth. Every prompt panel for a painter shares that basis and its limits. We additionally measure the ratio of separately centered sample covariance traces, using denominator $n - 1$ within each group. This descriptive ratio differs from the content-weighted population-form trace used in Study 1.

Fixed linear and radial-basis-function (RBF) kernel-ridge classifiers distinguish generated images from originals in all 31 coordinates, without using PCA scores or selecting hyperparameters.

Class weights are balanced and regularization is 0.01. The kernels are $K_{\text{lin}}(x, z) = x^\top z/31 + 1$ and $K_{\text{RBF}}(x, z) = \exp(-\|x - z\|^2/31) + 1$; the constant is a regularized intercept. Sixteen folds hold out one complete generated scene, including its four repeats, and a disjoint set of original work identities. Balanced accuracy averages the correct-classification rates for original and generated images; 0.5 is the chance reference. These splits do not hold out independent capture sources or service sessions.

To assess the finite-sample energy baseline, 128 deterministic splits divide each painter's originals into disjoint groups of size $n = \min(64, \lfloor N_p/2 \rfloor)$. Generated/reference comparisons use the same reference subsets and feature-independent generated subsets of size n. Each reported generated/reference median summarizes 768 distance estimates: 128 from each of six alias/method groups. Images are not pooled across groups. The same works recur across draws, so the resulting medians describe subsampling sensitivity rather than independent replications.

4.2 Distribution differences across all four painters

Figure 1 shows overlapping clouds with different locations and concentrations across all three formulations. The displayed PCs retain 44.1%, 44.4%, 47.8% and 42.9% of balanced variation for Monet, Sisley, Pissarro and Cézanne, respectively. More than half the variation lies outside these planes. The classifiers in Table 2 assess distinguishability in all 31 coordinates, including the variation omitted from the plots.

All four painters show lower generated spread and high classification accuracy (Table 2). The generated/reference trace ratios range from .206 to .376, and RBF balanced accuracy ranges from .940 to .983. The magnitude varies across painters and prompt formulations, while the direction of the spread difference is consistent across all 24 groups.

Painter	References	Trace ratio	Linear BA	RBF BA
Monet	297	.206–.304	.946–.973	.947–.982
Sisley	106	.219–.353	.947–.976	.970–.983
Pissarro	141	.240–.376	.934–.950	.940–.979
Cézanne	105	.249–.309	.913–.964	.942–.978

Table 2: Four-painter distribution summaries in all 31 features. Ranges cover all six alias/method cells per painter, each with 64 generated images; they are not uncertainty intervals. Trace ratios compare generated with original sample variance. Both fixed classifiers use held-scene/disjoint-work validation. Appendix B reports every cell.

Painter	Matched n	Original/original	Named/original	Free/original
Monet	64	.1932	2.0683	1.7319
Sisley	53	.2149	1.6776	1.8779
Pissarro	64	.1635	1.9745	1.8537
Cézanne	52	.1989	2.1419	1.6570

Table 3: Median all-feature energy discrepancy in matched-size subsets. Original/original medians summarize 128 disjoint-within-draw splits; each generated median summarizes 6×128 distance estimates. Works recur across draws; the medians are descriptive baselines, not equivalence thresholds or comparisons of independent capture sources.

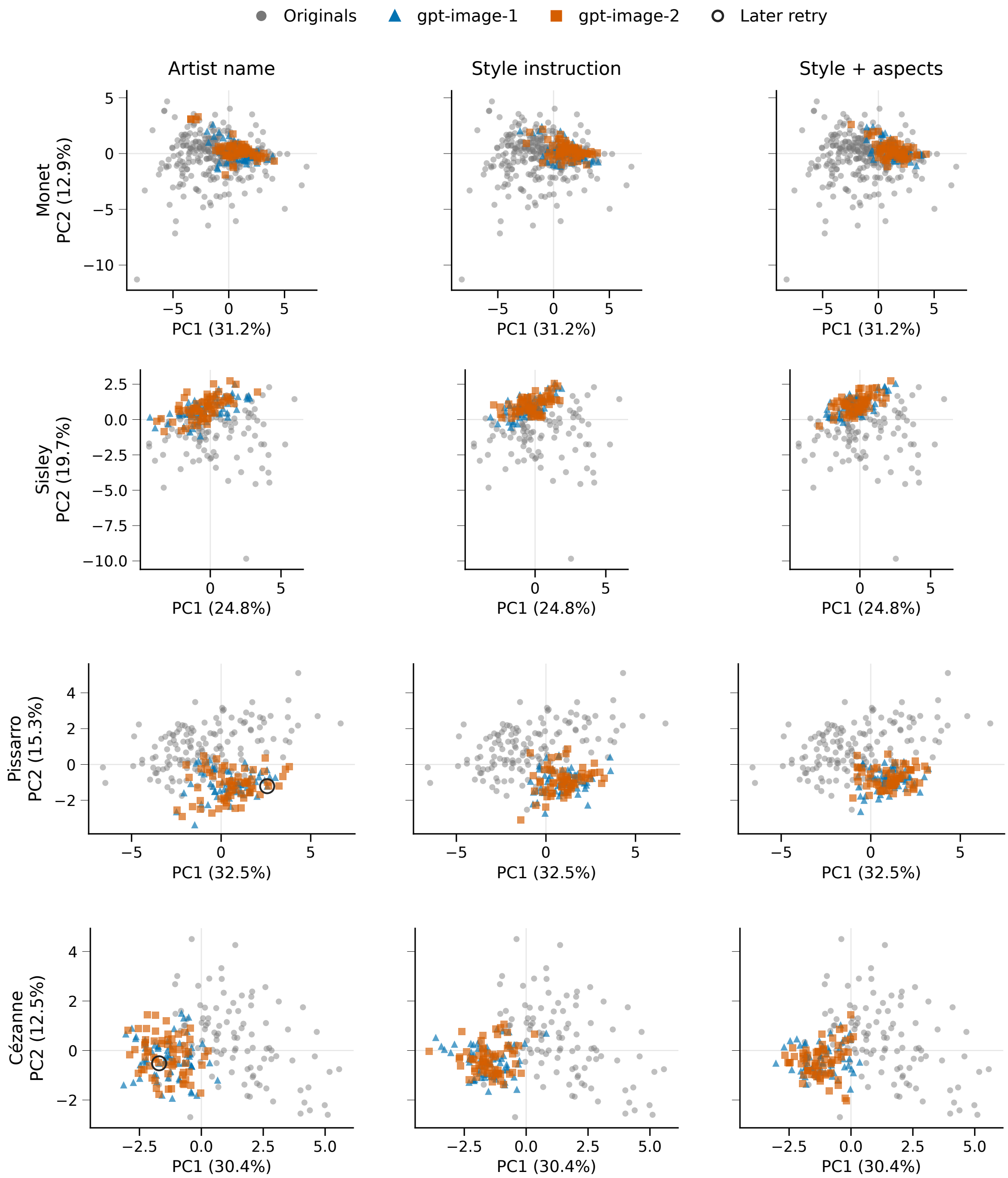

Figure 1: Original and generated distributions for all four painters and all three prompt methods. Gray points are originals; blue triangles and orange squares are the two requested service aliases. Each cell contains 64 outputs per alias. Black rings identify the two later retries. Originals are repeated across columns for comparison. All points and outliers are retained, with equal geometric aspect and common limits within each painter. Each row uses its own PCA basis; directions and coordinates cannot be compared across painters. The projection is descriptive and is not used by the classifiers.

Both generated/reference medians exceed the original/original median for every painter (Table 3), so unequal group sizes alone do not explain these energy gaps. The named median is lower than the artist-free median only for Sisley. In the separate feature-family prompt comparisons, by-name prompts have lower energy than explicit style instructions in all 24 alias–painter–family comparisons; adding aspect instructions reduces it in 13 of 24. These directional counts summarize correlated descriptive comparisons. Artist-free outputs also have high RBF balanced accuracy (.915–.990) and generated/reference trace ratios .369–.750, so discrimination does not isolate painter-specific mismatch.

The measured gap is compatible with content and capture differences as well as generation behavior. Equalizing four coarse content categories leaves named population-trace ratios at .211–.372, but categories describe recorded works and requested scenes, not independently verified content matches. The exploratory comparison therefore identifies differences among the observed image collections. Study 1 tests a narrower prompt intervention with a separate reference panel and common scene descriptions.

5 Controlled naming and distributional proximity

5.1 Design and reference panels

We compare 38 Monet and 32 Cézanne works represented by Wikimedia-delivered digital reproductions. Monet and Cézanne were selected after the four-painter exploration to prioritize contrasting observed patterns, additional service families and reference curation within a fixed collection budget. The reference works and generated images form a separate cohort; neither is pooled with the exploration. Reference identities were screened against prior project exposure. Availability and landscape eligibility determine this selected panel, rather than probability sampling of an oeuvre. Three coarse classes have water/built/land counts of 21/4/13 for Monet and 3/11/18 for Cézanne. A maintainer-run LLM coded these classes without independent human adjudication; 17 mixed or uncertain annotations remain flagged.

All detailed prompts begin "Create an oil painting on canvas." Naming adds "In the style of [painter]." before the otherwise identical scene and closing instructions. Eight scenes per class each have three repeats. No original image is supplied. The short-scene named condition on one service changes description specificity as well as length; both versions name the painter.

We request Nano Banana 2 (NB2; `gemini-3.1-flash-image`) and FLUX (`flux.2-max`) through fixed OpenRouter providers, and `gpt-image-2` through a local OAuth service. These identify requested services, not immutable weights. Conditions are randomly ordered within each service/painter/scene/repetition group. Collection spans 11.5 hours on 6–7 September 2026 UTC. The 1,008 assigned slots yield 1,006 images; two Cézanne short-scene requests are refused. All successful outputs are retained. Returned geometry and quality are part of the service outcome: all 576 NB2/FLUX outputs are square, whereas all references are nonsquare; OAuth rendering varies by condition. Appendix A gives prompts, delivery details and missingness accounting. Generated classes denote requested content, not verified adherence.

5.2 Analysis

Weighting and spread. References have equal weight, $a_i = 1/N_p$. If painter p has N_{pc} reference works in class c and a generated cell has n_{pc} available images in that class, a generated image receives $b_j = N_{pc}/(N_p n_{pc})$. Thus both distributions have the reference panel's class proportions. This weighting reduces differences in broad content mixture but leaves within-class content differences unresolved.

To measure spread, we use the weighted population-form covariance trace

$$\mathcal{V}(Y) = \sum_j b_j \|y_j - \bar{y}\|_2^2, \qquad \bar{y} = \sum_j b_j y_j. \tag{2}$$

We distinguish the generated/reference ratio $\mathcal{V}(Y)/\mathcal{V}(X)$ from the named/artist-free ratio $\mathcal{V}(Y_\mathrm{N})/\mathcal{V}(Y_\mathrm{F})$. The first measures spread relative to paintings; the second measures the change under naming. Sums of squared coordinate-wise IQRs provide a complementary summary.

For description b with total weight w_b and weighted mean $\bar{y}_b$, observed spread decomposes exactly as

$$\mathcal{V}(Y) = \underbrace{\sum_b \sum_{j \in b} b_j \|y_j - \bar{y}_b\|_2^2}_{W:\ \text{within descriptions}} + \underbrace{\sum_b w_b \|\bar{y}_b - \bar{y}\|_2^2}_{B:\ \text{between description means}}. \tag{3}$$

The observed term B also separates into within-class and between-class components. It contains noise from the three-repeat scene means; Section 7 gives the subsequent finite-repeat correction and its error assumptions.

Primary contrasts and inference. The prespecified inferential family comprises six named-minus-artist-free energy contrasts and two detailed-minus-short-scene named contrasts on OAuth. A contrast uses complete pairs sharing the description, repetition and assigned batch, with the same content weights in both conditions. The paired statistic retains all within- and between-collection terms in Equation 1. Appendix C gives the swap identity.

For each contrast, 99,999 Monte Carlo sign assignments yield a two-sided randomization p-value with the plus-one correction. Holm adjustment controls the fixed family of eight tests at level .05. The sharp null states that prompt assignment changes neither availability nor measured features in any assigned position, under the fixed request and technical-retry policy and no interference. For a pairwise OAuth comparison, the third condition's position is conditioned on. Inference is conditional on the assigned prompts and reference panel; it does not estimate an average effect over a painter's oeuvre or over new scenes. Treatment-dependent call duration or service carryover could violate no interference.

Study 1's primary tests and original descriptive analyses were fixed before generation. Table 1 distinguishes them from post-result sensitivities, alignment, matched-real coverage, decomposition, retrieval and cross-service detection, which add no confirmatory tests.

Alignment and distribution diagnostics. A relative alignment contrast compares both painter-conditioned collections G_M, G_C with both reference panels X_M, X_C:

$$I = \widehat{\mathcal{E}}(X_M, G_M) + \widehat{\mathcal{E}}(X_C, G_C) - \widehat{\mathcal{E}}(X_M, G_C) - \widehat{\mathcal{E}}(X_C, G_M). \tag{4}$$

All four distributions use equal one-third content-class masses, so panel mixture differences cannot by themselves produce the interaction. Within-distribution energy terms cancel; negative I favors joint own-painter pairing. Identical artist-free payloads under the two painter allocation labels provide a control for collection variation. This joint statistic does not require each named collection to be individually nearest its own panel.

Coverage is the fraction of reference neighborhoods containing a query (Naeem et al., 2020). In 100 fixed draws, generated and disjoint real queries share reference anchors and matched class counts, with neighborhood orders $k = 1, 3, 5$ (Appendix D.1). Appendix D.3 specifies processing, feature-view, reference-deletion and development-painter-omission checks.

5.3 Painter naming improves primary feature proximity

Painter naming lowers primary energy discrepancy in all six service–painter comparisons (Figure 2). NB2 decreases from 4.188 to 3.342 for Monet and from 4.704 to 2.886 for Cézanne; FLUX decreases from 1.990 to 1.365 and from 1.700 to 0.804, respectively. These four contrasts reject the conditional sharp null after adjustment across the eight-test family (Table 4). The smaller OAuth differences do not reject the null after adjustment, nor do the detailed-versus-short-scene comparisons. Non-rejection does not establish equivalence.

The six detailed named cells have generated/reference trace ratios .354–.756, compared with 1.069–1.535 for the four NB2/FLUX artist-free cells (Figure 2). Below-reference spread is therefore not shared by all generated conditions. Short-scene named OAuth ratios are lower than their detailed counterparts (.306 for Monet, .395 for Cézanne), although the detailed/short energy contrasts establish no advantage for detailed prompting.

Energy and spread are distinct but dependent summaries. For FLUX/Monet, twice the cross-domain mean distance falls from 15.831 to 12.421, while the within-generated mean distance falls from 6.906 to 4.120. Because the latter is subtracted in Equation 1, its decrease opposes rather than mechanically produces the observed energy improvement. The cross-domain change is larger. The coexistence of lower discrepancy and contraction is an empirical result here, not a necessary fidelity/diversity trade-off.

5.4 Relative painter alignment and reference coverage

Equal-class comparisons address whether the two names merely produce a common movement toward generic paintings. Naming makes the joint own/cross-painter interaction more negative on all three services (Table 5). The named-minus-free change remains negative across all 36 service/view/pipeline combinations. This joint relative contrast does not require every generated collection to be closest to its intended painter panel. In the primary equal-class comparison,

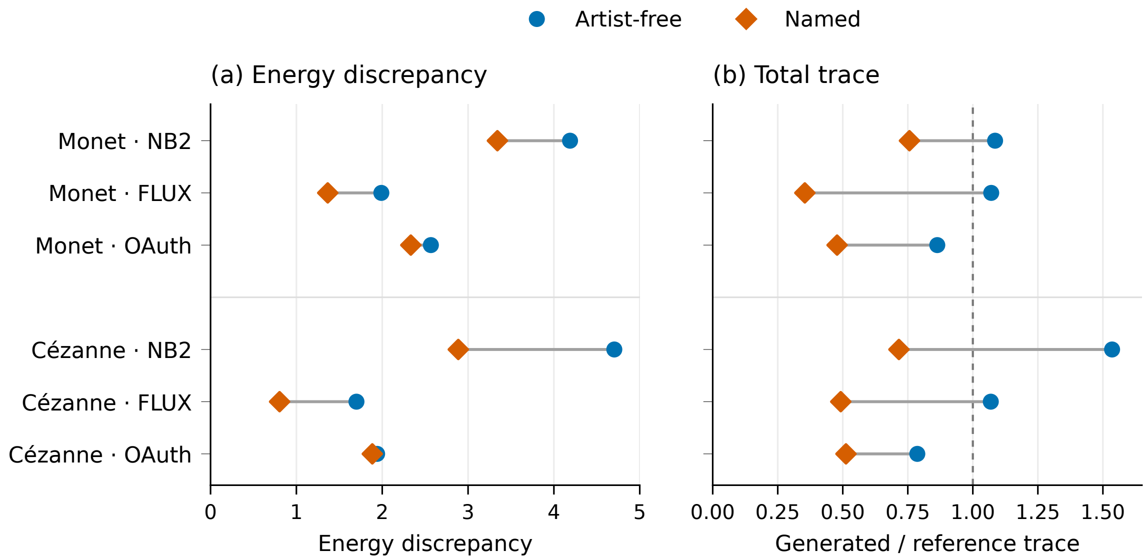

Figure 2: Primary feature discrepancy and spread under artist-free and named prompting. Energy compares each generated distribution with its painter's reference panel. Trace ratios compare generated spread with reference spread; the line at one denotes equal empirical trace. Each displayed generated cell contains 72 images; reference panels contain 38 Monet and 32 Cézanne works. The panels are descriptive and do not share a numerical scale.

Comparison	Service	Painter	Pairs	$\Delta\widehat{\mathcal{E}}$	Holm p
Named − free	NB2	Monet	72	−0.846	.00310
	NB2	Cézanne	72	−1.818	.00008
	FLUX	Monet	72	−0.625	.00008
	FLUX	Cézanne	72	−0.896	.00008
	OAuth	Monet	72	−0.234	.19308
	OAuth	Cézanne	72	−0.056	1.00000
Detailed − short scene	OAuth	Monet	72	−0.037	1.00000
	OAuth	Cézanne	70	−0.272	.09280

Table 4: The complete prespecified test family, using the primary 31-feature representation and matched pairs. Negative changes indicate smaller energy in the first condition. Values are conditional randomization results, with 99,999 sign draws and Holm correction across all eight comparisons. Both conditions in the final two rows name the painter.

NB2/Monet remains closer to the Cézanne panel (energy 3.488) than to Monet (3.687), despite improved joint alignment across the two named collections.

Service	Artist-free I	Named I	Named − free
NB2	.101	−.855	−.956
FLUX	−.048	−1.755	−1.707
OAuth	−.044	−1.802	−1.758

Table 5: Joint painter alignment under equal content-class weights in the primary feature space. Negative I favors own-painter over cross-painter pairing (Equation 4). Separately generated artist-free collections have identical prompt payloads across painter allocation labels. No additional significance test or equivalence threshold is assigned to this diagnostic.

At $k = 3$, naming increases median coverage in five service–painter cells and ties for OAuth/Monet, but every named median remains below the matched-real median (Table 6). Contraction can therefore accompany greater reference-neighborhood coverage. These fixed-panel comparisons remain descriptive.

Painter	Service	Artist-free	Named	Held-out real
Monet	NB2	0.278	0.500	0.944
Monet	FLUX	0.556	0.833	0.944
Monet	OAuth	0.500	0.500	0.944
Cézanne	NB2	0.200	0.400	0.933
Cézanne	FLUX	0.467	0.800	0.933
Cézanne	OAuth	0.467	0.533	0.933

Table 6: Median reference-neighborhood coverage at $k = 3$ across 100 matched finite-panel draws. Each draw uses the same original anchors for real and generated queries, with 18 queries for Monet and 15 for Cézanne. Generated query counts and content classes match the real queries. The medians describe the reused panel; they are not population confidence intervals.

Coverage depends on neighborhood size. At $k = 5$, FLUX/Cézanne reaches 1.000 for both named and real queries (Appendix D.1). This sensitivity precludes treating a coverage value as evidence of distribution equality.

5.5 Sensitivity to representation and references

All 72 named/artist-free total-trace ratios across the four feature views and three pipelines are below one, ranging from .308 to .816. Proximity is less consistent: 66 of 72 energy changes are negative. Both FLUX painter comparisons and NB2/Cézanne remain negative in all 12 view/pipeline combinations. For NB2/Monet, the 256-pixel color/spatial-only comparison instead increases energy by .416. The same no-texture view at 512 pixels gives a named/reference trace ratio of 1.101. Consequently, robust contraction relative to artist-free outputs does not imply robust below-reference spread or uniformly improved proximity.

The representation places substantial weight on texture. Texture coordinates contribute 49.0% and 47.1% of primary reference trace for Monet and Cézanne. Some multiscale coordinates are strongly correlated in development data. The feature views assess dependence on this representation; they reuse the same images and do not provide independent evidence.

Deleting individual reference works, holding-collection groups or content classes preserves the direction of all four NB2/FLUX primary proximity changes. These directions also survive omission of any one development painter from scaling. Some OAuth changes cross zero under these perturbations. These checks show stability of the NB2/FLUX directions within the available panel, while leaving shared reproduction differences unresolved.

5.6 Temporal recurrence on the same templates

A separately specified follow-up generated 72 FLUX images: one output for each of the 24 scenes under artist-free, Monet and Cézanne prompts, with the 24 free outputs shared between painter comparisons. A simultaneous 192-image OAuth palette collection gives four primary endpoints in a new Holm family. Allocation and inference were fixed before generation; all 264 outputs returned in 64.3 minutes. Appendix I details this collection and its tests.

Both FLUX naming effects recur: energy falls from 2.266 to 1.571 for Monet ($\Delta = -.695$, adjusted $p = .00006$) and from 1.772 to .820 for Cézanne ($\Delta = -.952$, $p = .00004$). Free/named trace ratios relative to the references are .737/.343 and .751/.553. The fresh free controls are already less dispersed than the references, unlike the original FLUX controls. Repetition of the naming direction therefore does not reproduce every baseline property. These cohorts share templates, reference panels and investigators; changed sample counts and shared controls preclude interpreting magnitude differences as an isolated time effect. The later FLUX design has no within-scene repeat-noise estimate.

5.7 Prospective comparison with a generic clause

We compared "In the style of Paul Cezanne." with "In a traditional landscape-painting style." on 24 new fixed scene briefs, eight per content class. Scene wording and the common opening/closing sentences are otherwise identical, without palette instructions. OAuth supplies two fresh outputs per scene and clause. This 96-output comparison was fixed after an availability failure in a four-arm allocation, before extracting features from that collection; its 211 returned images remain separate, with both original primary comparisons unavailable (Appendix A.3).

Each two-request block randomized clause positions and completed before the next. The comparison uses 48 pairs, the original 32 Cézanne references, reference weight 1/32 and query weight $q_c/16$. We test Cézanne-minus-generic energy using the 99,999-draw paired randomization procedure in Appendix C. Its sharp joint availability/feature null and no-interference assumptions must hold conditional on the prior availability-triggered decision. The raw-p threshold remains .025 despite the unavailable original endpoints; no further replacement, effect interval or equivalence claim is supplied.

All 96 images returned in 45.5 minutes without failures or retries. Naming lowers energy from 2.166 to .970 ($\Delta = -1.195$, raw $p = .00001$), rejecting the specified sharp null at $\alpha = .025$. This is the minimum attainable Monte Carlo p for 99,999 draws. The named collection also has lower observed spread: its trace is .552 times the generic trace and .362 times the reference trace. Both directions persist at 256 pixels and after JPEG 90 encoding (Table 7); these are descriptive views of the same images.

Pipeline	Generic energy	Cézanne energy	Difference	Cézanne/generic trace
Primary 512	2.166	.970	−1.195	.552
256 pixels	2.097	.632	−1.465	.719
JPEG 90, 512	2.161	.938	−1.223	.542

Table 7: Fresh generic-clause comparison: 48 images per arm, 24 new fixed scenes and 32 retained Cézanne references. All rows use 31 coordinates and unchanged content weights. Only the first row has a primary test; lower energy and trace do not establish perceptual similarity or distribution equality.

This tests two specific clauses on an authored panel, not a population of scenes or generic phrasings. All outputs are nonsquare PNGs; reported quality is medium for 16 generic and six Cézanne outputs and low otherwise, despite identical square/medium requests. The contrast includes these delivery differences. Changing scenes, controls, time and repeat count also prevents attributing its larger magnitude than the original OAuth contrast to the generic clause alone. The successor has no free arm and does not evaluate global feature maps or conditional predictions; prospective map validation therefore remains unavailable. The following diagnostics address the separate question of what reference proximity establishes about generated variation.

6 Can global feature transformations predict naming?

The following comparisons use different observational units and weights (Table 8); their numerical scales and sample counts are not interchangeable.

6.1 Generated-only fitting and whole-scene evaluation

We specified this diagnostic after observing Study 1 and the temporal FLUX results, before the prospective generic-clause follow-up. It uses all 864 detailed Study 1 outputs and the later 72 FLUX outputs, without new images or feature extraction. For generated training clouds F and N, let μ_F, μ_N be their weighted means and V_F, V_N their population-form traces. We compare identity with

$$T_1(y) = y + \mu_N - \mu_F, \tag{5}$$

$$T_2(y) = \mu_N + a(y - \mu_F), \qquad a = \sqrt{V_N/V_F} > 0. \tag{6}$$

The maps use the original content weights and no reference objective. The scalar matches total observed traces; it is not a noise-corrected latent response gain. Neither map rotates features or alters the relative geometry of scene centroids.

Each of four folds holds out six scenes, two per class; maps use the other 18 scenes and their three repeats. We score each 18-image held-out cloud separately and average across folds. Pooling different maps would add map-estimation variation. The fold score excludes between-fold generated distances, so its finite-sample baseline differs from the original 72-image energy; actual and transformed outputs share the fold scale.

The residual $\widehat{\mathcal{E}}(X, N_{test}) - \widehat{\mathcal{E}}(X, T_k(F_{test}))$ favors actual naming when negative. We report every fold and rerun the primary procedure after deleting each scene, preserving fold IDs and

Quantity	Observations compared	Weighting and aggregation
Full-cloud energy $\widehat{\mathcal{E}}$	72 images per detailed Study 1 arm; 24 per later FLUX arm; 48 per fresh Cézanne/generic arm	Each reference $1/N_p$; each generated class-c image q_{pc}/n_{pc}.
Mean held-scene energy	Four folds, each with 18 held images per arm (six scenes, three repeats)	Each reference $1/N_p$; each held image $q_{pc}/6$; equal mean of four separately scored folds.
Matched-real coverage	Monet/Cézanne: 18/15 anchors and 18/15 disjoint real or sampled generated queries per draw	Equal-weight anchor hits; matched class counts; median over 100 draws.
All-reference occupancy	All 38/32 reference anchors; 18 queries per held fold or 24 later queries, with six or eight per class	Equal-weight anchor hits, without query weights or resampling; equal fold mean for the original cohort.
Conditional residual Q_T	The same four scene folds; within each held scene, cross-products of distinct repeats ($R = 3$)	Scene mass $q_{pc}/2$ within each fold; equal fold mean; no painting-distance term.
Scene retrieval	72 held-repetition queries; 24 candidate centroids, or eight in the query's class; each centroid uses the other two repeats	Each query 1/72 (equal scene mass); accuracy is descriptive.

Table 8: Comparison units for the controlled panels and naming diagnostics. Here $N_p = 38/32$ for Monet/Cézanne, $q_{pc} = N_{pc}/N_p$ is the reference content-class proportion, and n_{pc} counts generated images of class c in the compared cloud. Counts describe complete cells, not independent replicates. Both neighborhood summaries use $k = 3$ in the main presentation; their anchors and query constructions differ. The matched-real allocation and other k values are in Appendix D.1.

restoring content masses within each fold (Appendix J). These overlapping diagnostics provide neither independent replications, equivalence conclusions, uncertainty intervals nor tests of a composite map model.

6.2 Held-scene performance

Actual naming has lower energy than translation/scaling in all six primary held-scene averages (Figure 3A). Residuals are $-.359, -.220$ for NB2, $-.312, -.192$ for FLUX and $-.129, -.026$ for OAuth, in Monet/Cézanne order. All four folds favor naming only for FLUX/Monet; the other cells contain folds with the opposite ordering. Delete-one-scene mean residuals remain negative in five cells, while OAuth/Cézanne spans $[-.122, .056]$. Actual naming also beats translation alone in five primary averages; OAuth/Cézanne is the exception. These outcomes describe predictive performance on the fixed scenes, without rejecting every possible location/scale account.

The residual against translation/scaling remains negative in all 60 specified metric and processing cells: the three original pipelines crossed with all 31 coordinates, omission of the blur-sensitive LBP coordinate and the 19 nontexture coordinates, plus the common-square all-feature view. Translation-only ordering is less stable, favoring naming in 51 of 60 cells. These are overlapping sensitivity results, not independent replications or a new inferential family.

6.3 What transfers to the later collection?

We fit one map per painter to all original FLUX observations and apply it unchanged to the later free outputs. Actual naming beats translation/scaling again, by .330 energy units for Monet and .172 for Cézanne. Translation alone, however, gives energies 1.470 and .736, below actual naming's 1.571 and .820. The translation advantage persists in 23 of 24 Monet scene deletions and all 24 Cézanne deletions. It holds in 15 of 18 processing/feature views; all three 256-pixel Monet views reverse the ordering.

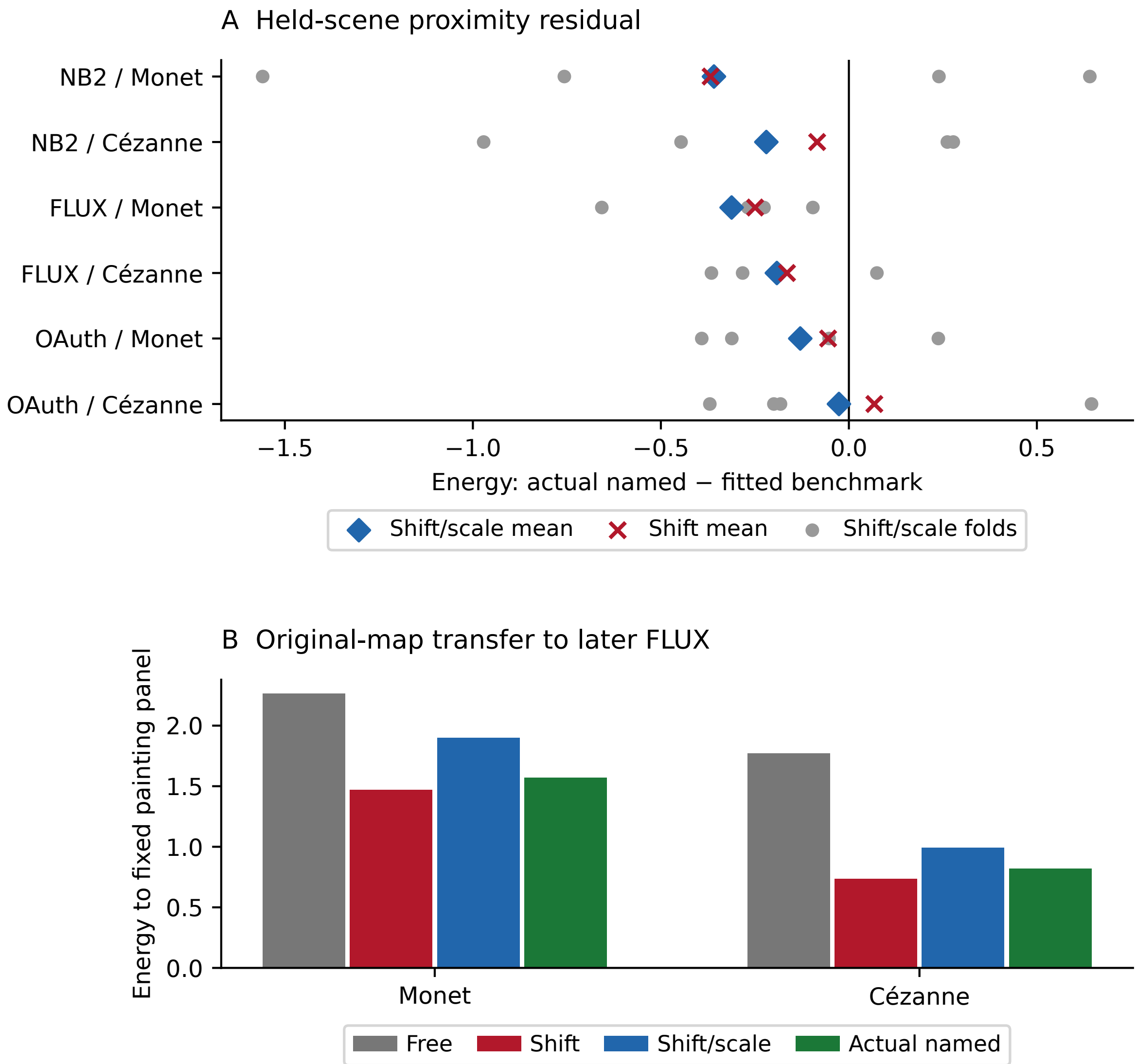

Figure 3: Testing a global moment account of naming. (A) Actual-minus-benchmark energy on held-out scenes: negative favors actual naming. Blue diamonds are means over four separately evaluated translation/scale maps; gray points show the four folds, and red crosses the translation-only means. Fold variation is not an uncertainty interval. (B) The original full-cohort FLUX maps applied unchanged to the later free outputs. Translation alone is closer to both reference panels than actual naming; translation/scaling is farther. Both panels use the primary 31-coordinate representation and reference-content energy weights. Panel A uses 18 queries per fold, panel B 24 per arm.

These outcomes do not isolate a cause inside FLUX. If the later free-cloud mean is m, the T_2 and T_1 output means differ by $(a-1)(m-\mu_F)$. Worse T_2 transfer therefore combines changes in center and spread; it does not identify the scalar's contribution.

6.4 Separating center choice from scalar contraction

To resolve this ambiguity, a separately specified post-result diagnostic adds one map using the evaluation-free mean m:

$$T_{2c}(y) = m + (\mu_N - \mu_F) + a(y - m). \tag{7}$$

It retains the training displacement and scalar. Its output mean equals that of translation, while its centered distances equal those of T_2. No evaluation- named or reference features set the new center. Unlike unchanged pointwise transfer, this map adapts to the observed free cohort; it supplies a controlled geometric comparison, not another prospective replication.

At this common mean, scaling increases energy in all 60 original and 18 later metric/processing cells. Later primary penalties are .591 for Monet and .254 for Cézanne (Table 9); the old center offsets .161 for Monet and changes Cézanne energy by only .002 along this path. Old-center anchoring therefore does not explain away the scalar penalty. The scalar matches original observed traces, without reference optimization; no optimal scalar or generation mechanism follows.

The energy terms explain this penalty. At the common mean, scaling reduces twice the mean reference/generated distance by 1.781 for Monet and 1.209 for Cézanne. It also reduces generated within-collection distance by 2.373 and 1.463. Because energy subtracts the latter term, the larger loss of within-cloud distance outweighs the improved cross-collection distance. These are exact decompositions of the observed contrasts, not estimates of a perceptual cost.

The difference between T_2 and translation decomposes algebraically as $[\widehat{\mathcal{E}}(X, T_{2c}F) - \widehat{\mathcal{E}}(X, T_1F)] + [\widehat{\mathcal{E}}(X, T_2F) - \widehat{\mathcal{E}}(X, T_{2c}F)]$. This is a specified comparison path, not a unique causal attribution. No new inference accompanies this comparison. Its evaluation mean couples residual repetitions, so we do not apply the independent-repeat conditional correction below to T_{2c}.

6.5 Coverage does not resolve the competing accounts

For the same clouds we count hits in fixed balls around every reference work, using its third-nearest other reference as the radius. Anchors, radii and paired query slots are shared across map comparisons. This all-reference occupancy uses the complete equal-class query sets in Table 8; unlike the matched-real coverage design, it has no real/real calibration.

In the later primary view, translation/scaling reaches .605/.750 of the Monet/Cézanne balls, compared with .526/.719 under actual naming, despite its larger energy discrepancies. Occupancy orderings vary across sensitivity views. The primary comparison is therefore a concrete disagreement between evaluators, not a representation-invariant coverage advantage. A hit records neither how much generated probability lies in a neighborhood nor agreement in condi-

Painter	Shift	Old-center scale	Evaluation-center scale	Actual named
Monet	1.470	1.901	2.062	1.571
Cézanne	.736	.992	.990	.820

Table 9: Later FLUX energy in the primary representation. The evaluation-centered map has the translation map's weighted mean and the old-center map's centered shape. Its worse score therefore isolates a scalar change at fixed mean in this specified feature comparison. All parameters except the evaluation-free mean come from the original cohort; no named evaluation outcomes are used for fitting.

tional scene structure. The result supports interpreting these counts alongside the underlying transformation, as motivated by Khayatkhoei and AbdAlmageed (2023).

7 Conditional organization after naming

7.1 Variation within and between scenes

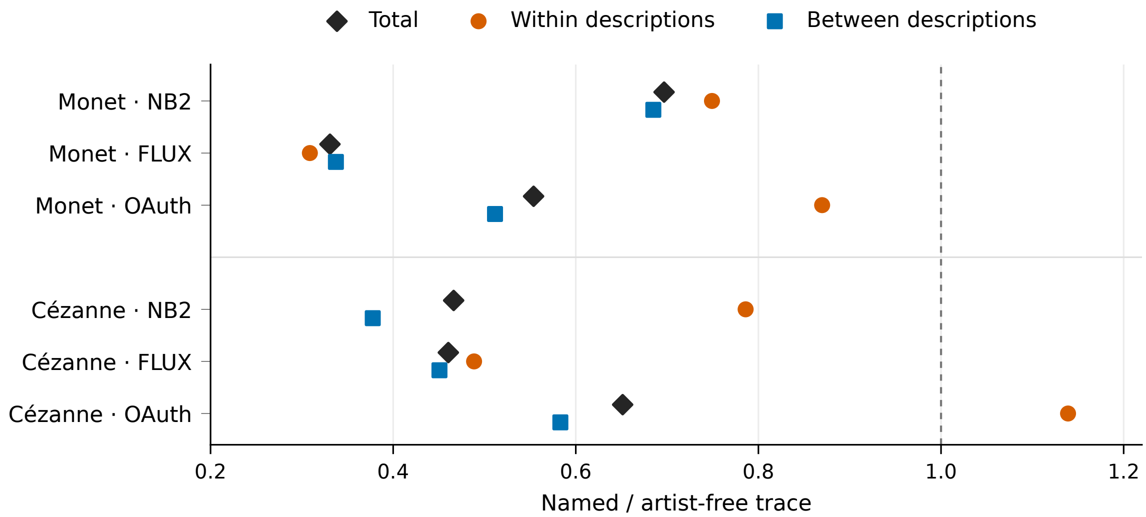

Figure 4: Where generated variation changes under naming. Each point is a named/artist-free trace ratio in the primary representation. Total variation is the sum of within-description and between-description components, so its ratio need not equal their unweighted average. The line at one denotes no change. OAuth/Cézanne contracts in total while increasing within-description variation. These finite-set decompositions are descriptive.

Primary named/artist-free total-trace ratios are .331–.697 (Figure 4). FLUX within-/between-description ratios are .309/.337 for Monet and .489/.451 for Cézanne. NB2/Cézanne's .786/.378 ratios show a larger relative change between descriptions than between repetitions.

OAuth/Cézanne illustrates a different pattern: total trace falls to .651 while within-description trace rises to 1.139 and between-description trace falls to .583. A tighter overall cloud can therefore coexist with more variable repetitions; aggregate spread alone conceals these distinct changes.

7.2 Separating scene means from repeat noise

The observed between-scene component includes noise in three-repeat means. For scene weights w_b and unbiased within-scene covariance S_b, a subsequent correction is

$$N^* = \sum_b w_b \,\mathrm{tr}(S_b), \qquad B^* = B - \frac{1}{R}\sum_b w_b(1-w_b)\,\mathrm{tr}(S_b), \quad R = 3. \tag{8}$$

Under stable zero-mean errors independent across repeats and scenes, B^* estimates variation among the fixed conditional means and N^* estimates repeat noise. Randomized request order does not ensure these assumptions. Negative corrected estimates are retained, and no population interval is introduced.

Both weightings preserve contraction of corrected scene variance (Table 10). Reference-content and equal-scene summaries differ: FLUX/Monet's named/free ratio of B^*/N^* is 1.107 under the former and .937 under the latter. The retrieval analysis below weights scenes equally, so the latter is its relevant scalar comparison. Even after correction, its retrieval gain cannot be attributed to an increased aggregate between-scene/repeat-noise ratio. Scalar traces leave the orientation and local arrangement of scenes unspecified.

For held-out residuals $e_{br} = N_{br} - T_2(F_{br})$, averaging $e_{br}^{\top} e_{bs}$ over distinct repeats removes the expected diagonal noise contribution (Appendix J). All six averages over folds are positive (Table 11), although one NB2/Monet fold is negative. Measured in squared standardized coordinates, these residuals locate conditional-mean mismatch not measured by marginal energy; they are neither significance nor equivalence tests.

A direct comparison of the two prediction targets is more revealing than the residual magnitude alone. Adding scale reduces the corrected conditional-mean prediction residual in every primary cell, yet increases reference energy in five (Table 11). Better prediction of the actual named scene means therefore need not improve proximity to the painting panel. This comparison concerns two specified fitted predictors; the correction removes held-out repeat noise, not training-parameter estimation error or all possible model misspecification.

7.3 Contraction need not reduce scene distinguishability

We classify each held-out Study 1 detailed named or artist-free repetition by the nearest scene centroid in Euclidean distance, computed from the other two repetitions. Cycling through all three repetitions gives 72 queries per service/painter/condition, using all 24 candidate scenes or the eight in the query's class. The latter removes discrimination solely from broad class. Development scaling remains fixed.

		Reference-content weights		Equal scene weights	
Service	Painter	B_N^*/B_F^*	N_N^*/N_F^*	B_N^*/B_F^*	N_N^*/N_F^*
NB2	Monet	.677	.749	.719	.815
	Cézanne	.316	.786	.360	.755
FLUX	Monet	.342	.309	.333	.355
	Cézanne	.443	.489	.448	.535
OAuth	Monet	.488	.870	.492	.838
	Cézanne	.544	1.139	.514	.980

Table 10: Named/free ratios after finite-repeat correction in the primary representation. B^* is corrected variation among fixed scene means; N^* is repeat-noise trace. Distinct class weights change the summaries, including the OAuth/Cézanne within-scene ordering. These are estimates conditional on the stated error assumptions, not tests or universal contraction coefficients.

		Conditional-mean residual Q_T		Reference energy	
Service	Painter	Shift	Shift/scale	Shift	Shift/scale
NB2	Monet	3.772	2.545	4.127	4.118
	Cézanne	6.908	2.794	3.277	3.412
FLUX	Monet	8.799	3.286	1.899	1.962
	Cézanne	4.243	2.063	1.219	1.246
OAuth	Monet	5.254	2.546	2.794	2.868
	Cézanne	8.354	6.224	2.122	2.216

Table 11: Predicting the named conditional means and approaching the references favor different maps. Both scores are lower-is-better, but have different units and targets. Entries average four held-scene folds in the primary metric. Q_T subtracts the repeat-noise contribution under its stated assumptions. The joint ordering occurs in 15 of 24 folds: adding scale lowers Q_T in 22 and raises energy in 16. All comparisons are descriptive.

We cross three pipelines with five views: all 31 features, color, spatial, texture and the 19 nontexture coordinates. Queries and variance comparisons use equal scene weights, unlike Equation 3. Overlapping centroid fits make accuracies descriptive, not 72 independent Bernoulli trials. Under $y' = ay + t$, $a > 0$, both variance components scale by a^2 and retrieval assignments stay fixed. Retrieval measures relative scene distinguishability rather than absolute response gain or requested-object presence.

In the primary 31-feature retrieval diagnostic, naming improves accuracy in two of six service–painter cells and lowers it in four (Figure 5). FLUX/Monet improves from 44.4% to 59.7%, although its equal-scene between-description trace falls to .336 of the artist-free value. Its retrieval gain remains positive under all three pipelines, including within-class candidate restriction. This example shows that lower between-scene spread can coexist with better retrieval in the observed collection. Transfer to new scenes or service runs remains untested.

The changes are heterogeneous. For Cézanne, accuracy declines from 66.7% to 41.7% on NB2 and from 83.3% to 69.4% on OAuth. Both declines persist across all five feature views and three pipelines in both candidate settings. For Monet, the OAuth decline is smaller within class (84.7% to 83.3%) than across all 24 scenes (79.2% to 70.8%). The NB2 difference is one correct query out of 72. Changes in separability therefore vary across services and painters.

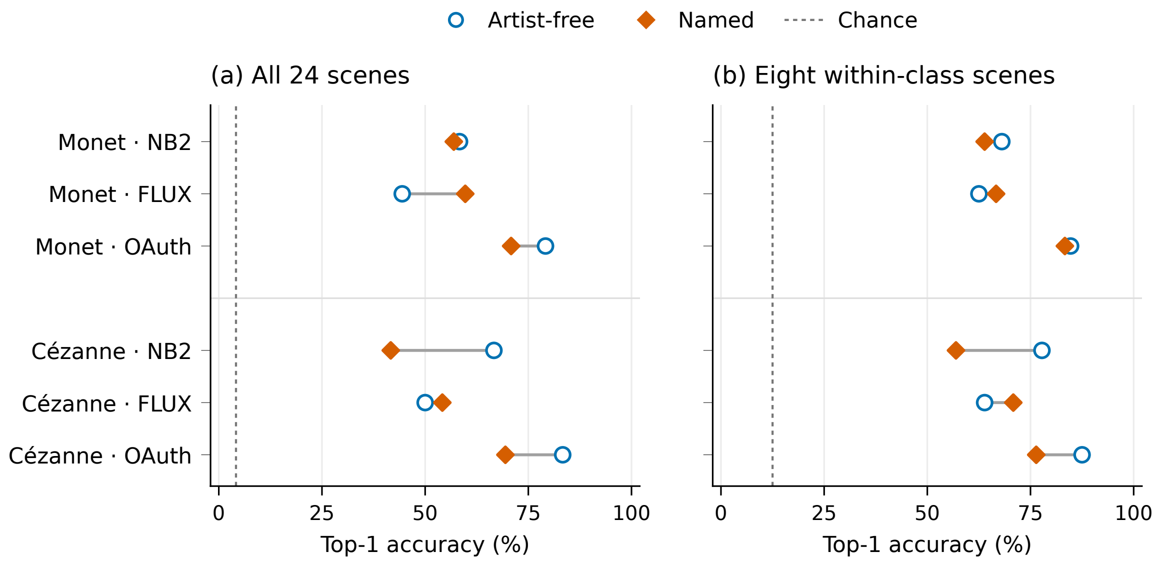

Figure 5: Held-repetition scene retrieval in Study 1. Each condition has 72 queries; centroids use only the other two repetitions. Panels compare 24 scene candidates with eight candidates in the same broad class. All scenes have equal query weight. Dashed lines mark chance rates of 1/24 and 1/8, respectively. These are descriptive accuracies in the primary 31-feature representation, not semantic-adherence scores or independent scene-population estimates.

8 Response to palette instructions

A separate experiment asks whether painter clauses attenuate response to an explicit palette instruction beyond a generic painting clause. Six fixed scenes are each generated under artist-free, generic traditional-landscape, Monet and Cézanne clauses, with muted/vivid palette instructions and four repetitions: 192 outputs per collection. The endpoint is median CIELAB chroma in development- IQR units. If δ_{bsr} is vivid-minus-muted response for scene b, style s and repeat r, the two primary interactions are

$$\kappa_M = \frac{1}{6}\sum_b \mathbb{E}_r[\delta_{bMr} - \delta_{bGr}], \qquad \kappa_C = \frac{1}{6}\sum_b \mathbb{E}_r[\delta_{bCr} - \delta_{bGr}]. \tag{9}$$

Here G is the generic clause. Shared controls enter the template-stratified covariance estimator. Welch–Satterthwaite approximations give simultaneous 95% family intervals and Holm tests for the two original interactions. They assume independent repeat-block errors and stable responses within the six fixed scenes; random request order does not make these model-based tests exact. The design, inference formulas and prospective proxy calibration are reported in Appendix E. The earlier 49-image incomplete collection remains ancillary and is not pooled with the sole primary run.

Both original interactions are unresolved: Monet $-.284$ with interval $[-.585, .016]$ and adjusted $p = .0649$; Cézanne $-.018$ with interval $[-.343, .308]$ and $p = .8886$ (Figure 6). The 24 scene/repeat blocks, not the 192 images, determine interaction precision. All six Monet scene estimates are negative; Cézanne has three of each sign. These intervals allow additional effects of either sign and do not establish equivalence with G.

The later, separately analyzed 192-output collection also leaves both interactions unresolved: Monet $-.159$ with interval $[-.475, .158]$ and adjusted $p = .349$; Cézanne $.037$ with interval $[-.263, .336]$ and $p = .727$. These intervals have nominal marginal coverage 98.75% within the new four- endpoint family that also contains the FLUX naming tests. Different interval levels and overlapping estimates do not test equality across collections.

The secondary generic-minus-free response is negative in both collections: $-.853$ with nominal 95% interval $[-1.136, -.570]$, then $-.891$ with $[-1.078, -.704]$. The generic clause itself therefore changes the measured response; an additional painter-name effect remains uncertain. The two palette extremes cannot distinguish a response-gain change from an operating-range shift or saturation. Descriptive chroma distributions relative to the painting panels appear in

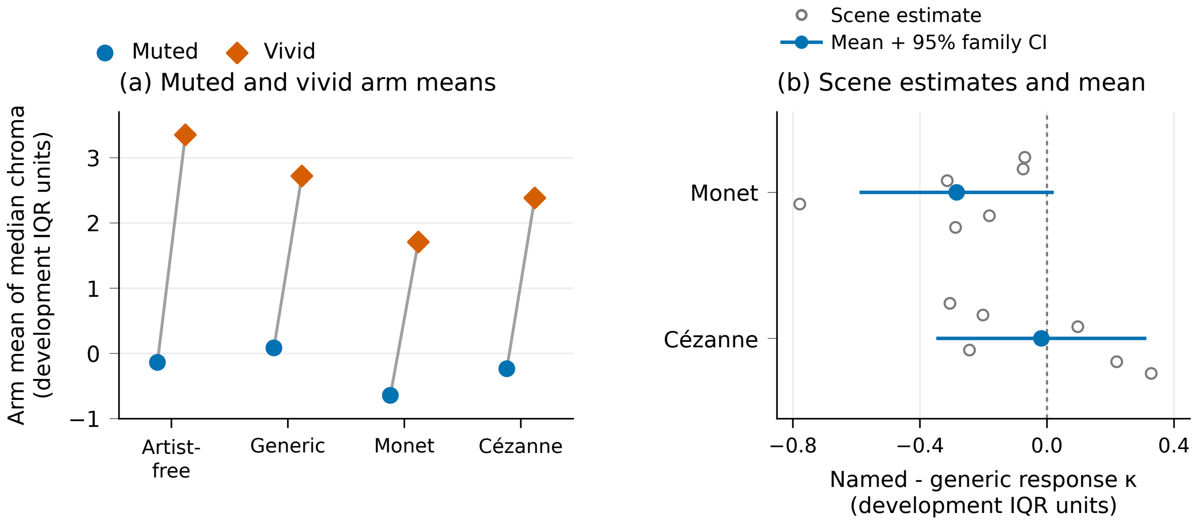

Figure 6: Study 2 color levels and additional painter-name interactions. Left: equal-scene mean image-median chroma, with 24 images per arm/polarity; these points have no uncertainty bars. Right: named-minus-generic differences in the vivid-minus-muted response. Gray hollow circles show all six scene estimates per painter; vertical offsets only prevent overlap. Blue points and bars show the equal-scene estimate and approximate simultaneous 95% family interval (Bonferroni marginal coverage 97.5%). Both panels use the same fixed chroma IQR unit but display different estimands. The two interactions share generic draws.

Appendix F. Neither these comparisons nor more repeats at the same two levels identify the cause of Study 1's contraction.

9 Measurement and processing boundaries

Computational challenges apply ten fixed conditions to all 70 reference works: a central-square baseline, lossless PNG, two JPEG qualities, resampling, two chroma contractions, two blur doses and tile permutation. The protocol fixes transformations before extraction; each comparison remains paired within work. The family response D_{iF} is root-mean-square displacement in the unchanged development-IQR coordinates. Appendix H gives the complete parameters, comparisons and work-resampling procedure.

All lossless round trips preserve pixels and features. The three specified change-minus-processing contrasts are positive: .158 for color, .305 for spatial features and 3.945 for texture, with descriptive intervals $[.139, .178]$, $[.239, .373]$ and $[3.680, 4.187]$. Responses are also non-selective and processing- dependent. One-pixel blur changes spatial features more than tile permutation; resampling produces a texture displacement of 1.099 versus .102 for JPEG 95. A post-result coordinate decomposition attributes 83.9% of the mean squared texture blur response to eight-neighbor LBP entropy. These observations motivate the moment-map sensitivity excluding that coordinate, without establishing a perceptual weighting or the capture origin of the generated/reference gap.

Common-square views preserve all eight original energy-contrast signs and the same four Holm rejection decisions (Appendix G). Median retained reference area is 75.0% for Monet and 78.8% for Cézanne; paid generated outputs were already square. The sensitivity therefore changes visible reference content as well as matching the window. It neither establishes independent-capture equivalence nor validates the original full-view domain classifiers. OAuth geometry and reported quality also vary by prompt condition and remain part of the delivered-service estimand.

10 Discussion

10.1 What the proximity gain establishes

The four-painter exploration finds differences between generated collections and digital references; the separate controlled intervention shows that naming can reduce them. The map comparisons test what this proximity gain establishes about generated variation.

Reference proximity and conditional prediction favor different maps (Table 11): adding scale reduces corrected mismatch to named scene means in all six primary cells, yet worsens reference energy in five. Holding the evaluation means equal retains the scalar's proximity penalty (Table 9), so center choice alone does not account for it.

The cohort comparison provides a complementary constraint. Actual naming beats translation/scaling in the six primary held-scene averages, whereas unchanged translation alone is closer than actual naming in the later primary FLUX view (Figure 3). Lower discrepancy therefore does not uniquely favor the named collection's variation. Transformed coordinates need not be feasible paintings, and their performance does not identify the service's internal computation.

Coverage and scene retrieval further separate these interpretations. A map can hit more reference neighborhoods while having worse energy. Uniform translation and positive scaling preserve retrieval, whereas actual naming changes it in both directions. Corrected between-scene variance contracts, but equal-scene scalar signal/noise does not explain FLUX/Monet's retrieval improvement. These observed disagreements apply established conditional-evaluation principles (Benny et al., 2021) to the painting comparisons.

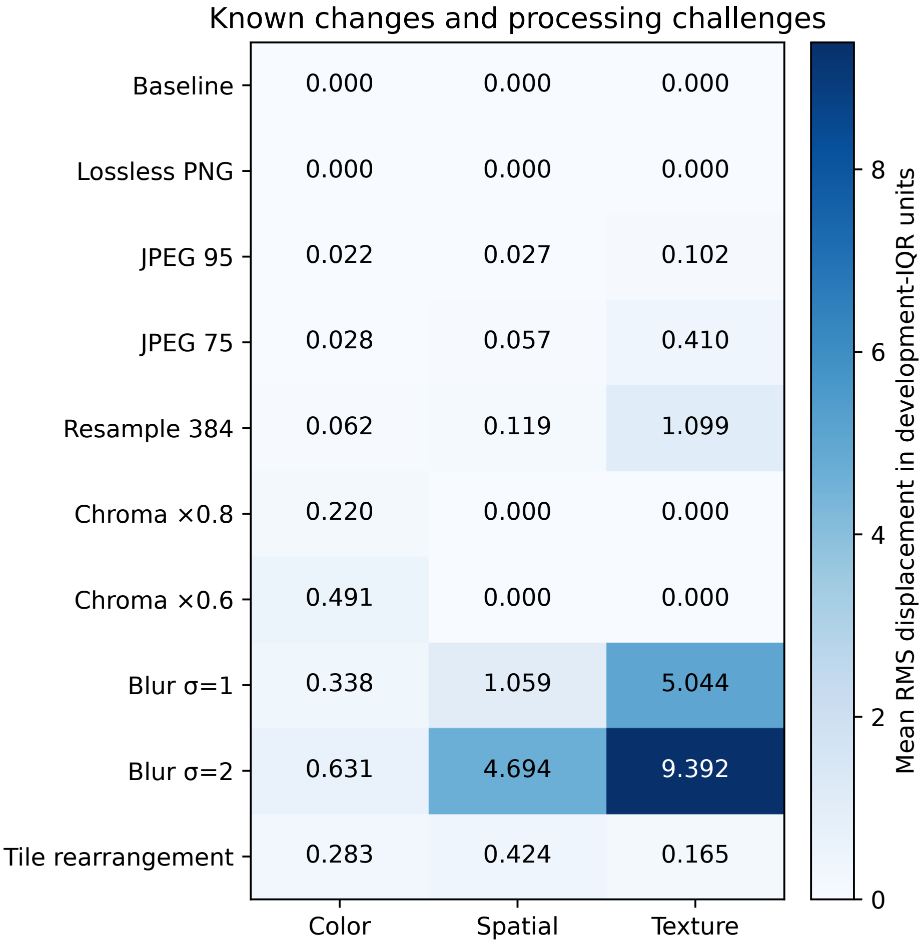

Figure 7: Feature responses to ten fixed conditions on the 70 Study 1 reference works. Cells show the equally weighted average of the two painter means of $D_{iF}(T)$ (Equation 13), including all cross-family responses. Each transformation starts from the same central-square baseline; derivatives are paired within work. Values rounded to .000 need not be exactly zero except for the baseline and lossless PNG. The common color scale describes numerical displacement, not perceptual importance or construct selectivity.

The new-scene Cézanne result shows lower discrepancy and spread relative to an actual generic painting clause, extending the artist-free comparison to one specified alternative. It does not establish that painter naming outperforms generic instructions in general, or that its effect is invariant across scene panels and service runs.

Neither palette collection resolves additional named-clause attenuation beyond the generic clause, although both secondary generic-minus-free responses are negative. This control dependence limits interpretation: chroma response under two explicit extremes cannot identify the cause of the broader distributional changes.

10.2 Scope and unresolved explanations

The features measure digital color, spatial organization and texture; their Euclidean weighting is not a validated perceptual scale. Unknown photographic workflows, correlated coordinates and the large texture contribution leave the origin of the discrepancy unresolved. Common-square and processing probes bound specific sensitivities but do not identify capture effects or physical brushwork. The numerical release enables replay from measurements, not public feature re-extraction from all pixels.

Painter and scene selection also limit inference. Sisley and Pissarro remain in the four-painter exploration, where only Sisley's pooled named energy median is below its free control. The later Monet/Cézanne contrasts use different panels, prompts and services and do not supersede that ordering. The two painters were selected after exploration; broad classes are maintainer-LLM annotations and requested scene content is not independently verified adherence. The prospective Cézanne/generic comparison uses new scene briefs, but the map and retrieval diagnostics reuse the original templates. Their transfer to new scenes or artists remains untested.

The original and temporal primary families were specified before collection; the map and variance diagnostics followed observed results. Their fold and scene-deletion stability does not give calibrated population uncertainty. Service carryover could violate primary randomization assumptions, while repeat correction and palette intervals require stable independent errors. All collections and LLM reviews were operated by the same maintainer. No independent investigator replication, human style validation or learned-feature comparison has been performed.

A next discriminating study would test fixed maps on newly specified scene panels and independently captured reproductions of the same works. Intermediate palette levels and several generic clauses would distinguish response slopes from shifts or saturation. These additions address specific remaining explanations; more images from the current templates alone would not resolve them.

11 Conclusion

Against artist-free controls, painter naming improves measured proximity in the original controlled Monet/Cézanne comparisons, with both FLUX effects recurring later. A new-scene Cézanne comparison also improves proximity against a generic painting clause while reducing spread. Held-scene and temporal transformation comparisons qualify its interpretation: actual naming outperforms global translation/scaling in all six primary held-scene averages, but a fixed translation alone is closer to both painting panels in the later primary view. Matching the evaluation means leaves a proximity penalty for the fitted scalar throughout the specified feature and processing views. Repeat-corrected contraction, heterogeneous retrieval and evaluator disagreements show that proximity is not a substitute for conditional organization. Together with the retained Monet, Sisley, Pissarro and Cézanne exploration, these results characterize prompt-dependent differences from digital painting distributions without claiming perceptual equivalence or an identified generative mechanism.

Data and code availability

The preceding versioned numerical reproduction package is provided at `https://github.com/isingmodel/latent-art-bench/releases/tag/pprv1-20260910`. It contains the manuscript, source code, locked dependencies, retained feature vectors, scalers, prompts, analysis memberships and numerical reports. A source catalog records reference-work identities, source URLs and recorded license metadata. The reproduction command recomputes the original statistical functions for the four-painter exploration, both original controlled studies, subsequent diagnostics, measurement challenges and temporal follow-up, then checks report values and all eight figures. Exact file hashes identify the released inputs; Appendix K specifies the replay scope.

The additive numerical package at `https://github.com/isingmodel/latent-art-bench/releases/tag/ppgv1-20260910` reproduces the subsequent analyses in Sections 6–7. It contains `painter_naming_geometry_v1` (run `pngv1-20260910`) and `painter_naming_centering_v1` (run `pncv1-20260910`), including their protocols, compact measurements, source bindings and complete results. The centering successor retains the geometry study's 60 original and 18 transfer cells and verifies its results exactly. These analyses use retained vectors and generate no new images. They supplement the unchanged preceding archive.

A further addendum at https://github.com/isingmodel/latent-art-bench/releases/tag/pcrv1-20260910 replays the generic-clause follow-up in Section 5.7. The original run, `pcvv1-20260910`, is in `painter_clause_validation_v1`; the fresh run, `pcsv1-20260910`, is in `painter_clause_successor_v1`. The archive retains the original unavailable endpoints, every allocated measurement status and the fresh cohort's complete results. Protocols, exact prompts, source bindings and locked dependencies accompany the compact vectors; no observations are pooled.

These releases support computation from retained measurements. Raw artwork, generated pixels and provider-response bodies are not redistributed, so the packages do not provide public feature re-extraction or independent authentication of the image acquisition. Recorded delivery fields and integrity-audit summaries are identified separately from recomputed statistical results. Source URLs do not guarantee current pixel access or permission to redistribute the underlying images. The code and author-produced numerical exports use the repository's MIT license, without granting rights to absent artwork.

A Features, prompts and collection details

A.1 Feature inventory

Family	Count	Coordinates
Color	11	Lightness median/IQR; chroma median/IQR; chromatic fraction; hue concentration/entropy; chromatic differences at three scales and their slope.
Spatial	8	Spectral slope/residual/anisotropy; orientation entropy; horizontal–vertical balance; gradient median/IQR; quadrant divergence.
Digital texture	12	Four wavelet energies, their slope and curvature; three local-binary-pattern entropies; local coefficients of variation at three scales.

Table 12: The fixed interpretable representation. Coordinates describe digital image statistics, not physical paint relief. The 28-coordinate view removes chromatic-difference slope, wavelet slope and wavelet curvature.

Color coordinates use CIELAB with D65 illumination and the 2° observer. Chromatic pixels satisfy $C^* \geq 5$; chroma-weighted hue concentration and 24-bin normalized hue entropy are set to zero when chromatic pixels occupy less than 1% of the image. Chromatic differences are median horizontal/vertical CIEDE2000 distances at lags of 1%, 4% and 16% of the short side; their summary slope is fitted in log–log coordinates.

Spatial and texture coordinates use linear-sRGB luminance. Spectral summaries use a Tukey window with parameter .1 and 32 logarithmic radial bins spanning 4–128 cycles per short side; the spectral slope uses a Theil–Sen fit. Gradient summaries use Scharr derivatives, an 18-bin magnitude-weighted orientation histogram and mean quadrant Jensen–Shannon divergence from the whole-image histogram. Texture coordinates comprise log mean squared detail energies at four levels of an undecimated `db2` wavelet transform, their slope and curvature, normalized uniform-LBP entropies at $(P, R) = (8, 1), (16, 2), (32, 4)$, and median local coefficients of variation at the three relative window sizes above. Exact boundary handling and numerical constants are specified in the versioned feature implementation identified in Appendix K.

A.2 Study 1 collection details

Reference eligibility screening excluded one figure-led work. The retained paintings are not matched to individual generated compositions.

NB2 and FLUX deliver 1024-by-1024 images in JPEG and PNG respectively. OAuth delivers PNGs with variable geometry: 424 of its 430 images report low quality and six report medium, all six in artist-free conditions. These delivery differences are retained in the prompt comparison. All three OAuth conditions are randomly ordered within each scene/repetition group. Repetitions are distributed across eight assigned collection batches, which do not denote independent backend sessions.

The service totals are 288 NB2, 288 FLUX and 430 OAuth images. Two Cézanne/OAuth short-scene requests are refused and remain missing. One transient FLUX failure is recovered by an exact-payload technical retry. All successful outputs are retained without selection for stylistic resemblance.

A.3 Generic-clause collection and availability

The initial new-scene design allocated 288 OAuth outputs: 24 briefs, three repetitions and four arms (artist-free, generic, Monet and Cézanne). One Cézanne request was refused; a later HTTP 500 response fell outside the fixed retry contract and stopped the collection. It retained 211 returned images, one refusal, one server error, one cancellation before posting and 74 unattempted slots. Its global collection restriction withholds both primary tests, and no arm has all 72 allocated images. Both primary comparisons and all complete-grid feature summaries are unavailable. All 211 returned images are nonsquare PNGs; 46 report medium quality and 165 low. None is selected for stylistic resemblance.

After the first refusal, before the later stop or feature extraction from this collection, we fixed one final 96-output Cézanne/generic successor. Its allocation, two repetitions and threshold .025 did not change when Monet also became unavailable. It retains every brief, including the refused scene, with both arms generated afresh. Original observations are neither pooled nor used to fill successor slots. Sources, allocation and request order were frozen before generation; at most two requests were active, recorded starts were at least five seconds apart and blocks drained before the next. All 96 outputs returned in 45.5 minutes and yielded measurements in all three pipelines. The successor provides the sole new-clause primary result reported in Section 5.7.

A.4 Exploratory prompt construction

Let S be the unchanged artist-free scene sentence, beginning “An oil painting on canvas of …”. The by-name condition inserts “by [painter]” immediately after “canvas”; its control is S. The explicit-style condition appends “Render this scene in the painting style of [painter].” to S, whereas its artist-free control appends “Render this scene in a coherent painting style.” The style-plus-aspects method adds the same final sentence to both of these conditions: “Attend to color relationships, the organization of forms, and the handling of painted marks; retain the stated scene.” Each method therefore has its own control wording. These contrasts differ from the single sentence addition tested in Study 1.

A.5 Controlled-study prompt construction

An example detailed named prompt is:

> Create an oil painting on canvas. In the style of Claude Monet. Scene: A pond viewed from close to its edge, with groups of floating lily pads, gaps of open water and reflections of surrounding trees. The water fills the scene; no horizon or prominent person is visible. Render only the painting area, without a surrounding frame, signature, letters or watermark.

The artist-free version removes only the painter-style sentence. The short-scene named version substitutes “A pond with water lilies.” for the detailed scene. All 24 descriptions are available in the accompanying prompt inventory.

Study 2 uses six separate fixed descriptions, covering a river and river bend, a village and houses, and fields and a garden. The common opening and closing instructions are retained. The style sentence is absent, generic ("In a traditional landscape-painting style."), Monet or Cézanne. The palette sentence is "Use a muted, low-chroma palette throughout the painting." or "Use a vivid, high-chroma palette throughout the painting." Each of the 48 scene/style/palette combinations has four outputs. The exact six scene texts and complete request inventory accompany the analysis.

B Complete four-painter distribution summaries

Table 13 reports all 24 cells underlying the main-text ranges. Aliases 1/2 denote the requested `gpt-image-1`/`gpt-image-2` routes; each cell has 64 generated images.

Painter	Alias	Prompt	Trace ratio	Linear BA	RBF BA
Monet	1	By name	0.249	0.946	0.953
Monet	1	Style	0.230	0.970	0.961
Monet	1	Style + aspects	0.227	0.973	0.980
Monet	2	By name	0.304	0.954	0.947
Monet	2	Style	0.206	0.955	0.982
Monet	2	Style + aspects	0.209	0.973	0.979
Sisley	1	By name	0.353	0.969	0.983
Sisley	1	Style	0.223	0.976	0.983
Sisley	1	Style + aspects	0.242	0.956	0.983
Sisley	2	By name	0.314	0.961	0.970
Sisley	2	Style	0.227	0.964	0.983
Sisley	2	Style + aspects	0.219	0.947	0.983
Pissarro	1	By name	0.342	0.934	0.940
Pissarro	1	Style	0.240	0.950	0.979
Pissarro	1	Style + aspects	0.240	0.946	0.967
Pissarro	2	By name	0.376	0.945	0.944
Pissarro	2	Style	0.267	0.946	0.963
Pissarro	2	Style + aspects	0.256	0.939	0.955
Cézanne	1	By name	0.309	0.951	0.978
Cézanne	1	Style	0.255	0.942	0.965
Cézanne	1	Style + aspects	0.262	0.942	0.948
Cézanne	2	By name	0.294	0.964	0.973
Cézanne	2	Style	0.254	0.913	0.942
Cézanne	2	Style + aspects	0.249	0.942	0.968

Table 13: All 24 four-painter cells in the full 31-feature space. Ratios use separately centered sample traces; both classifier scores pool out-of-fold predictions under the fixed 16-scene split. The entries are descriptive estimates without uncertainty intervals.

A four-fold sensitivity analysis grouped by assigned generation block gives by-name RBF balanced accuracies .986/.983 for Sisley and .960/.967 for Pissarro under aliases 1/2. These descriptive checks use the original block labels; the later retry images were not generated at those block times. Neither split evaluates transfer to independent generation sessions or capture sources.

C Paired randomization statistic

Let A_i, B_i be a retained pair with common weight b_i, and define $C(v) = \sum_j a_j \|v - x_j\|_2$ and $T(v) = \sum_i b_i(\|v - A_i\|_2 + \|v - B_i\|_2)$. Then

$$c_i = b_i\{2[C(B_i) - C(A_i)] - [T(B_i) - T(A_i)]\}, \qquad \sum_i c_i = \widehat{\mathcal{E}}(X, B) - \widehat{\mathcal{E}}(X, A). \tag{10}$$

Complementary pair swaps produce $\sum_i s_i c_i$, $s_i \in \{-1, 1\}$, exactly, including the within-generated distances. Independent randomized condition positions justify the sign assignments under the stated sharp null. Repeated scene descriptions are not treated as independently sampled scenes for population inference. With e sign draws at least as extreme in absolute value as the observed statistic, including ties, $p = (1 + e)/100{,}000$. All contrasts retain 24 distinct descriptions; the two missing short-scene outputs reduce only the Cézanne detailed/short-scene comparison to 70 pairs.

D Supporting Study 1 distribution diagnostics

D.1 Matched reference-neighborhood coverage

For each painter, each of 100 fixed draws splits the references into disjoint anchor and real-query groups within content class. The groups have 18 works each for Monet (water/built/land 10/2/6) and 15 each for Cézanne (1/5/9); unmatched odd works are omitted in that draw. Generated queries are sampled without replacement with the same class counts. Every anchor's radius is its distance to the kth nearest other anchor in the fixed 31-feature, primary-512 space. Coverage is the fraction of anchor balls containing at least one query. Real and generated queries share each draw's anchors and radii. The draws reuse the panel; medians and draw ranges do not estimate uncertainty across independently sampled painting populations.

Painter	k	NB2 named	FLUX named	OAuth named	Real
Monet	1	.222	.444	.278	.500
	3	.500	.833	.500	.944
	5	.667	.944	.722	1.000
Cézanne	1	.133	.467	.333	.533
	3	.400	.800	.533	.933
	5	.667	1.000	.733	1.000

Table 14: Median matched-anchor coverage across 100 finite-panel draws. Neighborhood size changes the numerical comparison; saturation at $k = 5$ for FLUX/Cézanne does not establish matching distributions. The accompanying numerical tables include artist-free conditions and all draw memberships.

D.2 Alignment controls

Equation 4 uses equal broad-class masses. Sparse reference strata therefore receive one-third weight, giving reference effective sample sizes of about 24.0 for Monet and 18.8 for Cézanne; the nominal counts are 38 and 32. The weighting changes the contribution of existing works; it does not add independent reference information. For all 216 matched artist-free pairs across painter labels, request payloads are verified identical. Their separate finite collections have V-energies .079, .102 and .077 for NB2, FLUX and OAuth. These values illustrate collection and finite-sample variation, without defining an equivalence margin or a significance threshold for alignment.

D.3 Processing, representation and reference sensitivities

We examine three processing pipelines: the primary 512-pixel pipeline, a 256-pixel version and common JPEG encoding at quality 90 with 4:2:0 subsampling at 512 pixels. Each uses its own development-only scaler. Four feature views cross these pipelines: all 31 coordinates; 28 coordinates after dropping three derived multiscale summaries; 31 coordinates divided by the square root of their family size; and the 19 color/spatial coordinates with texture removed. The family-rescaled squared distance sums mean squared differences within each family; it does not equalize observed family variance or perceptual importance. This gives 12 views per service–painter comparison.

Reference diagnostics delete one work, holding-collection membership group or content class at a time. Collection groups use exact recorded sets of holding- collection IDs and do not identify photographic capture workflows. Work and collection-group deletions preserve the original content-class masses while reweighting the remaining works within each class; class deletion renormalizes the surviving class masses. A separate analysis refits the scaler after omitting each development painter.

D.4 Projection and domain detection

For visualization, we fit one common principal-component analysis (PCA) per painter, assigning half the weight to references and sharing half equally across the seven generated cells, with content weighting within each. All cells use that painter's common axes. Full-space linear and radial-basis-function (RBF) kernel-ridge classifiers use six folds holding out entire generated descriptions and disjoint original works, without tuning or PCA inputs. This grouping respects the repeated-prompt structure (Roberts et al., 2017). Cross-service evaluation trains on one service and tests on another using the same held-out description/work scheme. We report balanced accuracy at a fixed zero decision threshold and mean within-fold area under the ROC curve (AUC) separately; the former also depends on calibration.

The PCA views show overlapping clouds with changes in location and concentration (Figure 8). The first two components retain 48.4% of weighted variation for Monet and 46.0% for Cézanne. With more than half the variation omitted, these projections provide only a partial view of the distributions.

In the full feature space, within-service RBF balanced accuracy ranges from .893 to .987 across the 14 original/generated comparisons. However, all 576 NB2/FLUX images are square and all 70 references are nonsquare: a rule using only square geometry separates these domains perfectly. Capture workflows for the reference reproductions are also unresolved. High classification accuracy therefore establishes differences between the recorded image domains, not a validated stylistic distinction.

Cross-service behavior further limits detector interpretation. At the fixed threshold, naming lowers balanced accuracy in every directed service/painter comparison for both kernels. Named-condition ranges are .410–.874 for the linear kernel and .447–.704 for RBF. This need not mean uniformly weaker ranking discrimination. For Monet with training on FLUX and testing on OAuth, linear balanced accuracy falls from .904 to .874 while mean within-fold AUC rises from .953 to .968. The observed threshold transfer includes calibration changes; below-chance scores are not evidence that generated images have become more authentic.

D.5 Classifier specification

For $d = 31$ standardized coordinates, the fixed kernels are $K_\mathrm{L}(u, v) = u^\top v/d + 1$ and $K_\mathrm{R}(u, v) = \exp(-\|u - v\|_2^2/d) + 1$. The classifiers minimize weighted squared error with a kernel penalty of .01, using equal total training weight for original and generated labels and the reference content

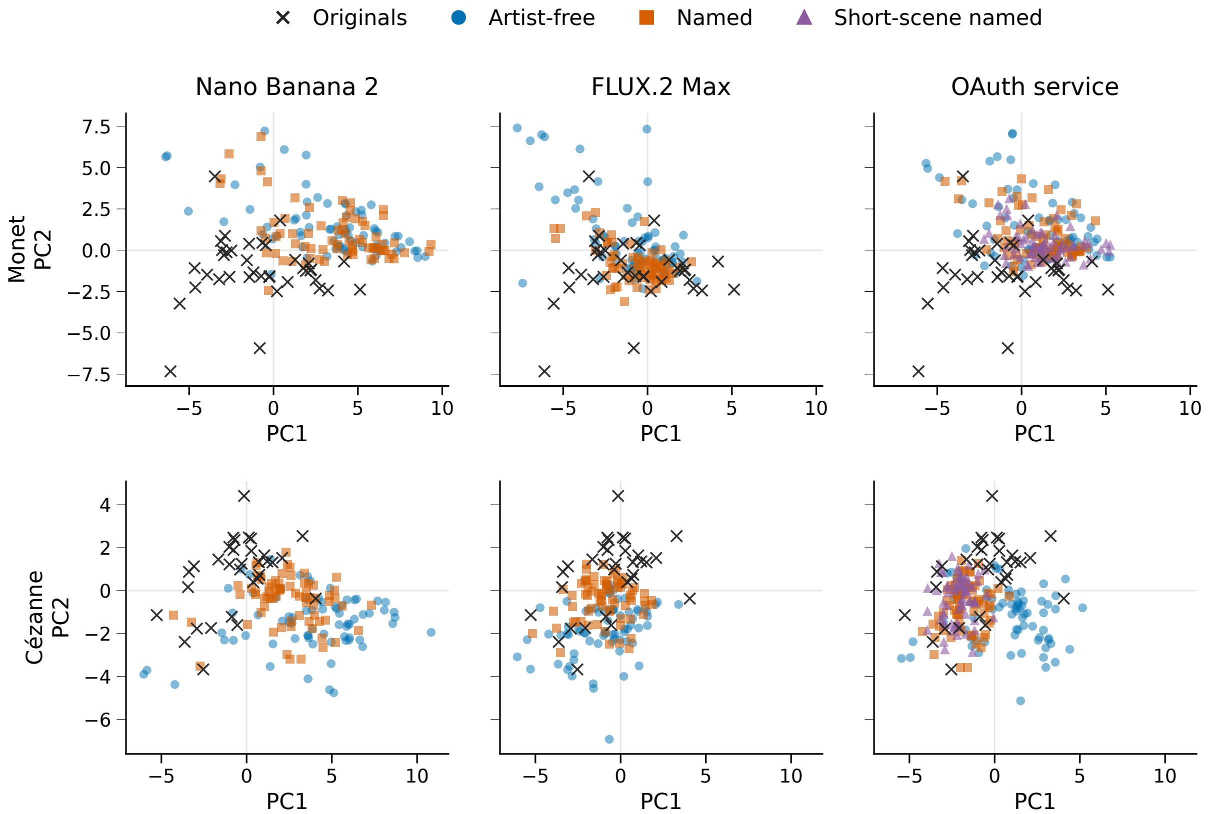

Figure 8: Original and generated feature distributions in common painter-specific PCA coordinates. Every Study 1 image is shown; references are repeated across service panels for comparison. Axes and limits are shared within each painter, but the two painters have separate PCA fits. The short-scene named condition is available only for OAuth. Projections show overlap and concentration while retaining less than half of total weighted variation.

proportions within each. Training and evaluation memberships are fixed. Sparse reference classes mean not every test fold contains all three classes; pooled balanced accuracy uses the complete held-out set. For cross-service evaluation, AUC is computed within each fold and averaged across six folds; margins from differently trained classifiers are not pooled. Complete kernels, memberships and scores are included with the analysis code and numeric tables.

E Color experiment: estimation and acquisition

For fixed scene $j = 1, \ldots, J$, repetition $r = 1, \ldots, R$ and arm a, let $D_{jra} = z_{jra,+} - z_{jra,-}$ and $K_{jr} = (D_{jrM} - D_{jrG}, D_{jrC} - D_{jrG})^\top$, with $J = 6$ and $R = 4$. Writing $\bar{K}_j = R^{-1} \sum_r K_{jr}$ and S_j for their within-scene sample covariance, the estimates and covariance are

$$\hat{\kappa} = \frac{1}{J} \sum_j \bar{K}_j, \qquad \widehat{\mathrm{Cov}}(\hat{\kappa}) = \sum_j \frac{S_j}{J^2 R}. \tag{11}$$

This retains the off-diagonal covariance from shared generic controls. For endpoint p, let $v_{jp} = S_{j,pp}/(J^2 R)$. The approximate degrees of freedom are

$$\nu_p = \frac{(\sum_j v_{jp})^2}{\sum_j v_{jp}^2/(R-1)}. \tag{12}$$

The two simultaneous intervals are $\hat{\kappa}_p \pm t_{1-.05/(2\cdot 2),\nu_p}\sqrt{\sum_j v_{jp}}$. Two-sided t p-values receive Holm adjustment within this two-endpoint family. The scenes are fixed; between-scene variation of mean responses is not added as a random-scene variance component. Estimated scene means retain generation noise. Request-order randomization does not establish the assumed independence and stability of repeat errors.

E.1 Repeat-block variation

Figure 9 is a post-result display of all 24 observed interactions for each painter. Monet has 17 negative block contrasts, although all six scene means are negative. The garden scene contributes 25.7% of Monet's estimated interaction variance; the fields scene contributes 44.5% of Cézanne's. These shares are $v_{jp}/\sum_j v_{jp}$ under the unchanged stratified estimator. They describe the concentration of estimated uncertainty within the six fixed scenes. No blocks are removed, and the display does not establish independence or stability across service collections.

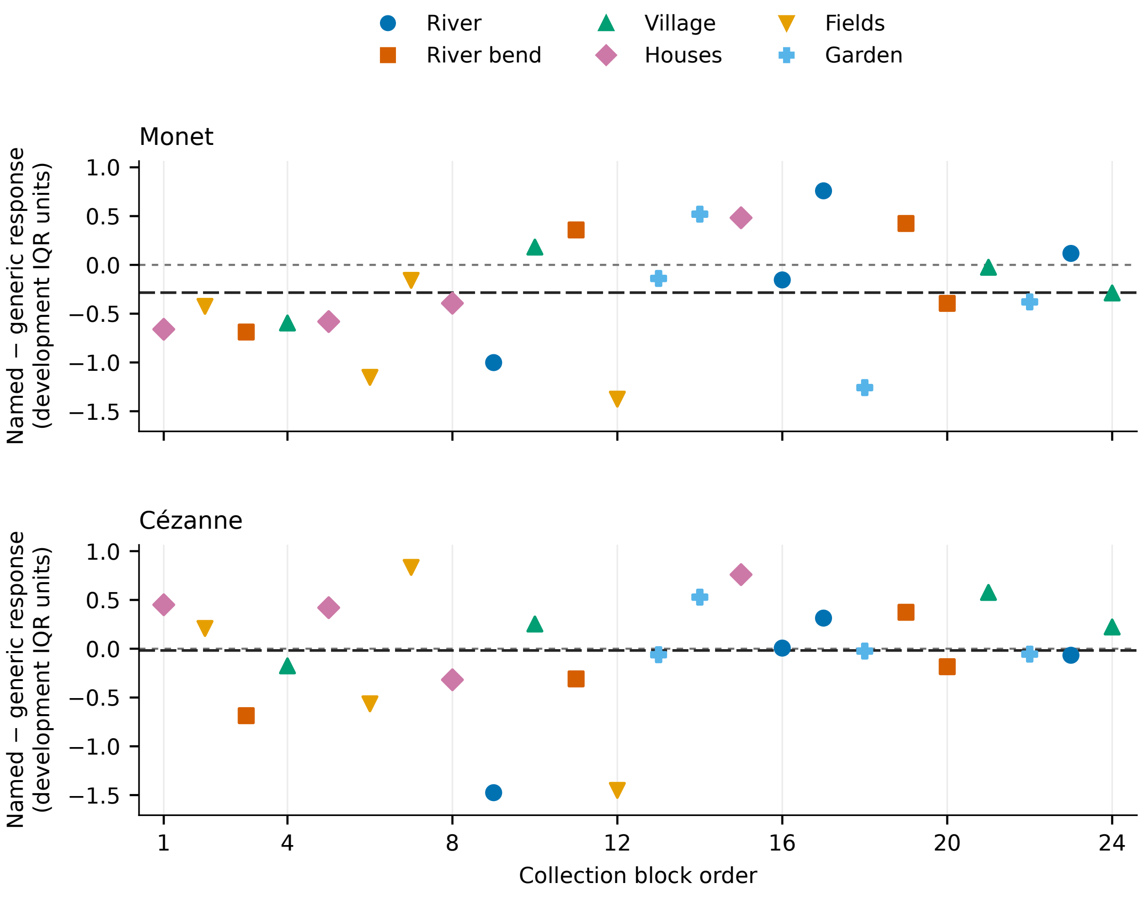

Figure 9: All repeat-block named-minus-generic chroma-response interactions in Study 2. Each point combines the named and shared generic vivid/muted outcomes from one eight-request block. Colors and symbols identify the six scenes; the horizontal axis gives collection-block order, which matched the fixed dispatch schedule. Both panels retain all 24 blocks on a common vertical scale. Reference lines mark zero and the pooled interaction estimate with short and long dashes, respectively. This descriptive display adds no significance test.

E.2 Prospective calibration and collection history

A prospective precision simulation used centered historical OAuth within-scene residuals with finite-repeat variance correction. The generic arm's noise was proxied by artist-free residuals, without validation that the two distributions match. Each of three independent-noise scenarios used 5,000 trials for each specified interaction setting. Besides the empirical proxy, a second scenario multiplied generic noise by 1.5. A heteroskedastic normal scenario applied scene standard-deviation multipliers from .7 to 1.8 to arm-specific baseline noise, then muted/vivid factors of .8/1.4, a generic factor of 1.5 and a Cézanne factor of 1.25. Table 15 reports the global-null rows. Null-family error over the partial-null settings ranges from .0336 to .0428. The hypothetical .25/.5/1-IQR interactions assess precision under specified effect sizes; they are not perceptual thresholds or post-hoc power estimates. These simulations support the procedure under complete-availability, independent proxy errors; they do not establish realized-service independence, tail behavior or coverage.

Noise scenario	Holm family rejection	Simultaneous interval coverage
Empirical proxy	.0382 [.0332, .0439]	.9618 [.9561, .9668]
Generic noise $\times 1.5$	.0330 [.0284, .0383]	.9670 [.9617, .9716]
Heteroskedastic normal	.0404 [.0353, .0462]	.9596 [.9538, .9647]

Table 15: Prospective global-null calibration in 5,000 trials per scenario. Brackets are 95% Wilson Monte Carlo intervals for simulation proportions, not confidence intervals for experimental effects. Each row uses independent simulated repeat errors and complete availability.

An earlier allocation was stopped after 49 images and one complete HTTP 503 response, leaving 142 assigned slots unstarted. A transport correction and a replacement collection using the same fixed prompts and order were specified before visual inspection or feature measurement of the successful images. The 192-image replacement is the sole Study 2 inferential cohort. The 49 earlier images were measured after replacement collection ended and remain ancillary, without pooling or filling missing slots. All 192 replacement requests succeeded, with no failures, retries or missing measurements. Every output was a nonsquare RGB PNG despite square requests. Reported quality was low for 174 and medium for 18 despite a medium-quality request; medium counts were 10, 4, 3 and 1 in the free, generic, Monet and Cézanne arms. No output was excluded or adjusted using these delivery fields. The effect therefore includes rendering differences associated with assignment; reported quality is a service field, not a measured quality score.

F Designed chroma mixtures and the reference panels

The named arms' equal-polarity mixtures differ from the reference panels in both mean chroma and the allocation of probability mass (Table 16). Most reference paintings lie inside the generated central range, while most generated outputs lie outside the reference central range. Broad range inclusion therefore does not establish a matched distribution.

Equal broad-class reference weighting gives named-mixture Wasserstein distances of .603 for Monet and .759 for Cézanne, compared with 1.071 and .787 for the same generic-control mixture. Naming lowers this distance in both cases, with directions preserved across all three pipelines and both reference weightings. The comparison shows closer chroma distributions under naming for these designed mixtures. The two extreme instructions do not establish continuous coverage of the intervening range.

	Mean chroma		Mass inside central range		
Painter	Original	Generated	Gen. in orig.	Orig. in gen.	W_1
Monet	.369	.533	37.5%	89.5%	.620
Cézanne	.812	1.075	16.7%	93.8%	.716

Table 16: Chroma distributions in Study 2 relative to the empirical reference panel. Generated values use each painter's 48 named outputs with equal class and polarity mass. Central ranges are empirical 10th–90th percentiles, not confidence intervals. Means and Wasserstein-1 distances use development-IQR units. The forced muted/vivid mixture is not an unprompted sampling distribution.

G Common-square energy contrasts

Table 17 reports all eight original contrasts after replacing aspect-preserving full views by their central 512-pixel squares. The scaler, reference/content weights, pairs, missingness, Monte Carlo seeds and eight-test Holm family are unchanged. These are subsequent window sensitivities, not new primary tests or independent image observations. The unchanged full-view calculation replays exactly before the new measurements are used.

Comparison	Service	Painter	Full view	Square	Square Holm p
Named − free	NB2	Monet	−.846	−.813	.00430
	NB2	Cézanne	−1.818	−1.797	.00008
	FLUX	Monet	−.625	−.532	.00008
	FLUX	Cézanne	−.896	−.906	.00008
	OAuth	Monet	−.234	−.244	.17259
	OAuth	Cézanne	−.056	−.110	.62136
Detailed − short	OAuth	Monet	−.037	−.043	.74574
	OAuth	Cézanne	−.272	−.279	.10312

Table 17: Energy-contrast sensitivity to a common central window. There are 72 pairs per row except the final row, which retains the same 70 complete pairs. All directions and adjusted rejection decisions agree with the full-view results in Table 4; agreement does not establish equality of effects.

H Computational measurement challenge details

H.1 Controlled changes to the measured images

We specified a computational challenge after the preceding analyses, fixing all transformations and comparisons before extracting the new measurements. It uses all 70 Study 1 reference works and the unchanged extractor and development scaler. After the original aspect-preserving normalization, we take the central 512×512 window as a common baseline. Each work contributes ten conditions: the baseline; a lossless PNG round trip; JPEG encoding at qualities 95 and 75 with 4:4:4 subsampling; Lanczos resampling from 512 to 384 pixels and back; Lab chroma contractions by factors .8 and .6; Gaussian blur in linear RGB with standard deviations 1 and 2 pixels; and a fixed permutation of sixteen 128×128 tiles. PNG must preserve pixels and features exactly, and tile permutation preserves the joint RGB pixel multiset. We retain gamut-clipping fractions for the chroma transformations.

For work i, feature family F and transformation T, the response is

$$D_{iF}(T) = \sqrt{\frac{1}{|F|} \sum_{j \in F} \left[\frac{f_j(Tx_i) - f_j(x_i)}{s_j} \right]^2}, \tag{13}$$

where s_j is the unchanged development IQR. Three principal paired comparisons subtract the larger within-work response to JPEG 95 or resampling from the response to .8 chroma contraction in the color family, tile permutation in the spatial family, and 1-pixel blur in the texture family. The processing challenges are not assumed perceptually irrelevant. We report both painter means and their equally weighted average. Descriptive percentile intervals resample whole works within painter 9,999 times, preserving all transformations together, at quantiles $.05/6$ and $1 - .05/6$. These panel-resampling intervals use a Bonferroni-three convention without establishing coverage for a painter population. Derivatives do not increase the number of works.

Separately, we apply the same central window to all 1,006 Study 1 generated images and recompute the eight original energy contrasts with unchanged pairing, weights, missingness and inference. This common-square sensitivity removes the gross shape difference while also removing visible content. It therefore tests dependence on the image window, not equivalence of photographic captures.

I Temporal collection and inference

I.1 Replication design

To test whether the observed findings recur in newly generated images, we specified a separate 264-output temporal follow-up. FLUX receives all 24 original detailed scene briefs once under each of three arms: artist-free, Monet and Cézanne. The 24 artist-free outputs are shared between the two painter comparisons. OAuth receives the original six palette scenes with four repetitions of the same four-arm, two-palette design, giving 192 outputs. Payloads, allocation, request order, inference and transport rules are fixed in a separate source and inventory freeze before generation. Within each three- or eight-request block, condition positions are randomized; the two experiments' blocks are then interleaved. At most two requests are active, with starts at least five seconds apart and the current block completed before the next begins. The paid allocation is smaller than Study 1's 72 outputs per painter and condition; it does not estimate variation among repetitions of a FLUX prompt.

The follow-up has four primary endpoints: two named-minus-free energy contrasts and two named-minus-generic chroma-response interactions. For each naming comparison, we condition on the other named arm's position and use the original paired-contribution randomization test. We report no confidence interval for the naming contrasts. Palette estimation retains the original shared controls and fixed-scene covariance estimator. Holm adjustment spans all four endpoints; the two palette intervals allocate $\alpha/4$ to each endpoint and have nominal marginal coverage 98.75%. These are conditional within-collection analyses, without a variance estimate across collection dates. They retain the palette model's assumptions of stable response distributions and independent repeat-block errors during the new interleaved collection. Zero estimated variance withholds inference for the affected palette endpoint. If elapsed collection time exceeds 24 hours, all four primary tests and the palette intervals are withheld, with observed estimates retained descriptively.

A negative naming contrast with rejection in this four-test family defines directional replication. The earlier palette interactions were unresolved, so there is no established attenuation effect to declare replicated. Old and new palette estimates are compared descriptively, without treating interval overlap as an equality test. Shared controls and changed output counts also preclude interpreting differences between the old and new energy estimates as an isolated time effect. A missing primary measurement withholds the affected experiment's allocated-grid inference; no later outputs fill its missing slots.

All 264 allocated images returned in 64.3 minutes, with no failures or retries. The 72 FLUX images were 1024×1024 PNGs. As in Study 2, all 192 OAuth images were nonsquare PNGs despite square requests. Reported quality was medium for 20 of 48 artist-free, 4 of 48 generic, 3 of 48 Monet and 4 of 48 Cézanne outputs, and low for the remaining 161. These delivery differences are retained in the prompt comparisons; reported quality is not a measured quality score.

A post-result audit finds no exact image duplicates among the 264 new outputs or between them and the 1,006 Study 1 and 192 primary Study 2 outputs, using both encoded-file hashes and primary normalized-array hashes with dimensions. This check does not exclude near-duplicates or establish independent backend states. The collection and its computational checks were performed by the same maintainer; this is temporal replication, not replication by independent investigators.

J Held-scene moment-map specification

Within each of water, built and land, the eight scene IDs are sorted and assigned to folds by zero-based index modulo four. Each fold holds out two scenes per class and all their repetitions. Training and evaluation weights separately restore reference-content masses. Reference feature vectors never enter the map fit; the fixed content masses come from the reference-panel composition. All maps require finite positive training traces.

Each fold is evaluated separately; averaging differs from the full-cloud energy because between-fold generated distances are absent. In the delete-one-scene sensitivity, fold IDs and equal fold averaging stay fixed. Restoring each fold's class mass gives a surviving scene in a depleted class more mass than same-class scenes in other folds. These are stability diagnostics, not globally equal-weight estimates on a new scene population.

For conditional-mean residuals, with $e_{br} = N_{br} - T(F_{br})$, define

$$Q_T = \sum_b w_b \frac{\sum_{r \neq s} e_{br}^\mathsf{T} e_{bs}}{R(R-1)} \tag{14}$$

$$= \sum_b w_b \left(\|\bar{e}_b\|^2 - \frac{\mathrm{tr}(S_{e,b})}{R} \right). \tag{15}$$

Conditional on one fold's fitted map, zero-mean errors independent across repeats and scenes make cross-repeat products estimate squared conditional-mean mismatch. Within-repeat free/named error dependence is permitted. Independent-partition products are an established way to remove positive noise bias from squared distances (Diedrichsen and Kriegeskorte, 2017); this calculation is Euclidean, without noise whitening. Conditioning jointly on overlapping maps does not establish independence across their residual estimates. Negative estimates are retained. Q_T sums squared changes over coordinates, so its units differ from the unsquared distances in

Comparison	Painter	Earlier	Fresh	Fresh Holm p	Fresh interval
FLUX named − free	Monet	−.625	−.695	.00006	—
	Cézanne	−.896	−.952	.00004	—
OAuth response interaction	Monet	−.284	−.159	.349	[−.475, .158]
	Cézanne	−.018	.037	.727	[−.263, .336]

Table 18: All four primary temporal-follow-up endpoints. Naming contrasts are energy differences; palette interactions are chroma-response differences in development-IQR units. Earlier naming comparisons contain 72 pairs, versus 24 fresh pairs with a shared free arm. Each palette collection has 24 eight-cell blocks. Fresh tests share one four-endpoint Holm family; the displayed palette intervals have nominal marginal coverage 98.75%. Earlier inferential families and palette interval levels differ. Cohorts are analyzed separately; magnitude differences are not a test of temporal change.

Service	Painter	Free	Shift	Shift/scale	Named	Residual	Delete-one range
NB2	Monet	4.872	4.127	4.118	3.759	−.359	[−.559, −.180]
	Cézanne	5.148	3.277	3.412	3.192	−.220	[−.376, −.138]
FLUX	Monet	2.627	1.899	1.962	1.650	−.312	[−.642, −.080]
	Cézanne	2.144	1.219	1.246	1.055	−.192	[−.275, −.142]
OAuth	Monet	3.185	2.794	2.868	2.739	−.129	[−.247, −.043]
	Cézanne	2.312	2.122	2.216	2.189	−.026	[−.122, .056]

Table 19: Primary held-scene energies and actual-minus-shift/scale residuals. Values average four separately fitted/evaluated folds. Delete-one ranges summarize 24 overlapping recalculations, not confidence intervals. These fold values must not replace the original 72-query energies in Table 4.

energy. It supplies neither a perceptual scale nor a test of complete conditional-distribution equality.

For translation/scaling, the naive weighted squared-mean residuals are 6.983, 6.026, 5.527, 3.973, 4.610 and 7.700, in the row order of Table 19. Subtracting repeat contributions gives 2.545, 2.794, 3.286, 2.063, 2.546 and 6.224. Positive raw residuals alone would therefore overstate conditional-mean mismatch. Twenty-three of 24 primary fold estimates are positive; the negative NB2/Monet fold is retained. Neither this sign count nor the six positive averages supplies a calibrated rejection of the transformation model.

The variance correction follows from $\mathbb{E}[B] = B_{\text{means}} + \sum_b w_b(1 - w_b)\,\text{tr}(\Sigma_b)/R$ under independent scene-mean errors. Replacing Σ_b by its unbiased sample covariance yields Equation 8. No truncation is applied to negative B^*. Synthetic known-map and heteroscedastic-error checks validate the implementation; they do not verify these assumptions for the delivered services.

All sensitivity cells retain the same folds and identities. The later-cohort analysis fits each map once on the original FLUX scenes and holds it fixed during the 24 later-scene deletions. It cannot estimate a new repeat variance with one image per scene/arm. The complete export includes fitted means/scalars, fold membership, energy terms, per-anchor occupancy, corrected per-scene contributions and every specified deletion result.

K Analysis files and reproduction

The public package records file checksums and original source provenance. The scientific implementations are copied unchanged; the release adapter supplies compact inputs in their original contracts. The feature extractor is `src/latent_art_bench/painter_feature_generation_v2/features.py`. The release README identifies the three original prompt inventories and their construction code. Reproduction requires neither API credentials nor the maintainer's Git history or response archive.

With Python 3.13.11 and locked dependencies, the `check` command in `tools/paper_release.py` recomputes energy tests, spread, PCA coordinates, grouped predictions, coverage, painter alignment, retrieval, palette interactions and quantile corrections. The measurement extension replays all transformation comparisons and common-square contrasts; the temporal extension replays its separate four-endpoint family and any completeness or duration restrictions. Retained seeds, memberships, weights and endpoint availability are unchanged. Direct comparisons connect recomputed values to the tables used by the figure builders.

The release README gives the complete command and platform settings. `COVERAGE.json` maps manuscript result families to recomputation or recorded-metadata status. Numerical replay uses the released inputs and installed runtime, with network and external-file access guarded. The portable comparison uses 10^{-10} absolute and relative floating-point tolerances, including identified continuous Welch p-values and their Holm transforms. Identities, counts, randomization and other p-values, and scientific decisions remain exact. The release README records the Welch-specific comparator amendment made after cross-platform verification exposed final-bit differences. Platform-dependent PDF bytes are reported separately from numerical agreement. The original protocols and feature implementation are available for inspection; source pixels are outside the public package.

The additive geometry archive supplies both complete versioned protocols and an independent replay route for Sections 6–7. After checking its archive checksum, verify the extracted inventory before installing the locked runtime:

```
python tools/paper_geometry_release.py verify --root .
uv sync --locked --python 3.13.11 --extra analysis --extra dev
uv run --locked python -m latent_art_bench.painter_naming_geometry_v1 verify
uv run --locked python -m latent_art_bench.painter_naming_centering_v1 verify
```

The first module verifies `pngv1-20260910`; the second verifies `pncv1-20260910` and its unchanged predecessor results. Both recompute from the compact numerical inputs and compare against sealed outputs. The manifest and accompanying SHA256 list bind every file, including the protocols, source, inputs, results and nine figure PDFs. This addendum does not rerun the historical studies; those require the preceding archive and its separate command.

The clause archive has its own supported numerical adapter:

```
python tools/paper_clause_release.py verify --root .
uv sync --locked --python 3.13.11 --extra analysis --extra dev
uv run --locked python tools/paper_clause_release.py check --root .
```

The first command checks the inventory using only the standard library; the last recomputes both clause cohorts with their unchanged eligibility rules, assigned weights and seeds. Original unavailable outcomes remain unavailable. This separate package neither invokes the generation workflow nor checks the other two releases. Its README lists the included numerical and package tests.

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